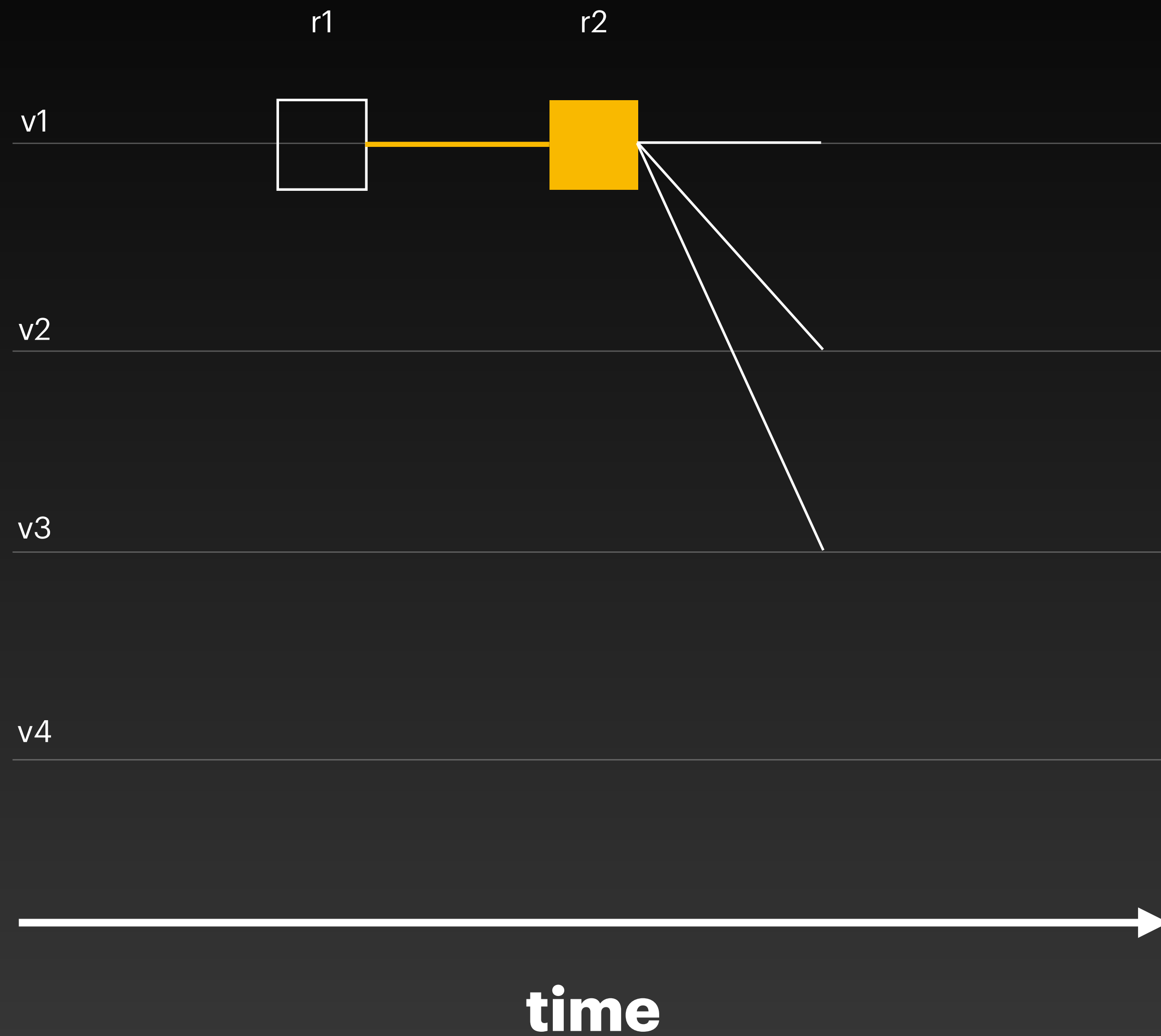


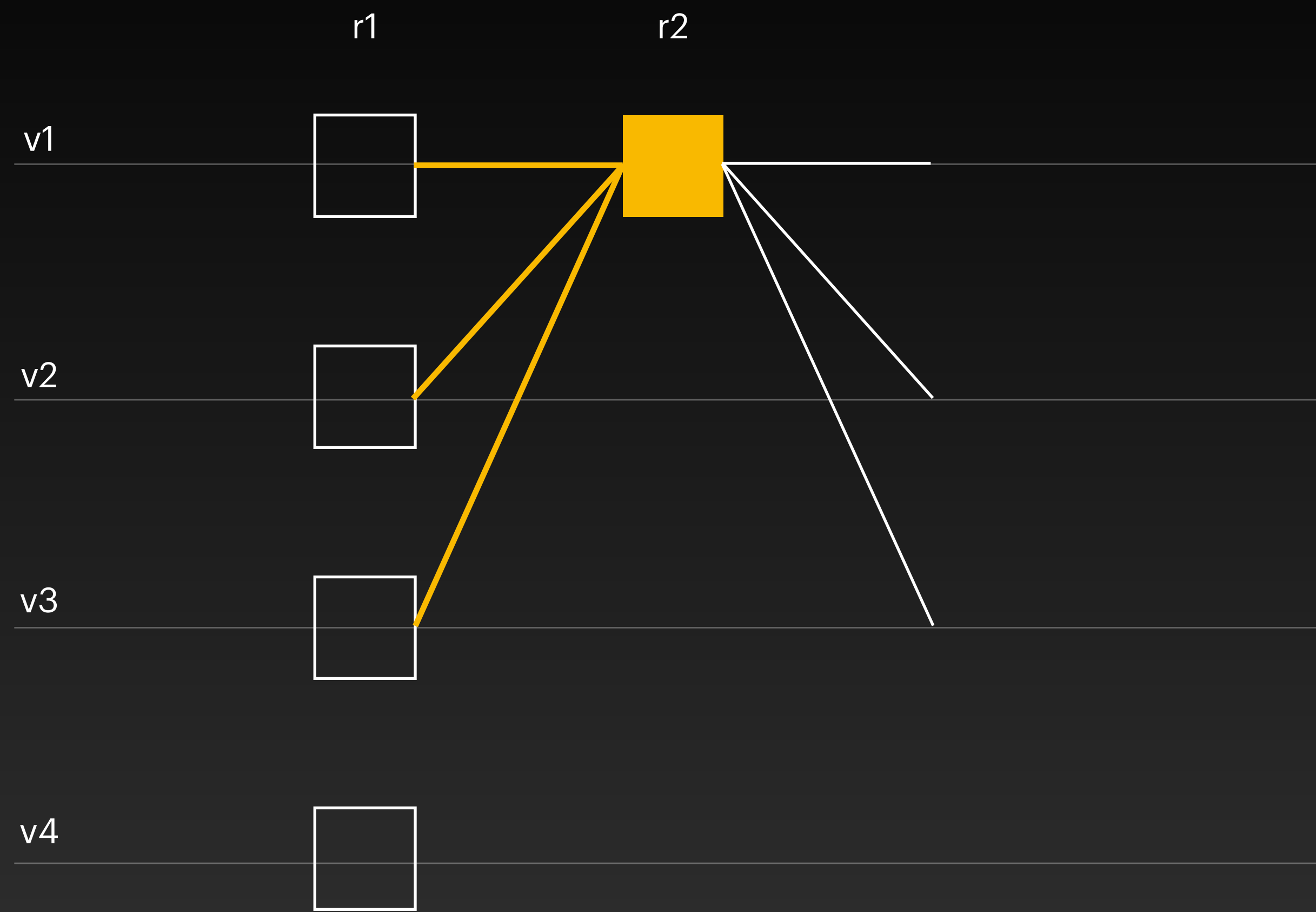
One DAG to Rule them All

Hydrozoan

Alberto Sonnino

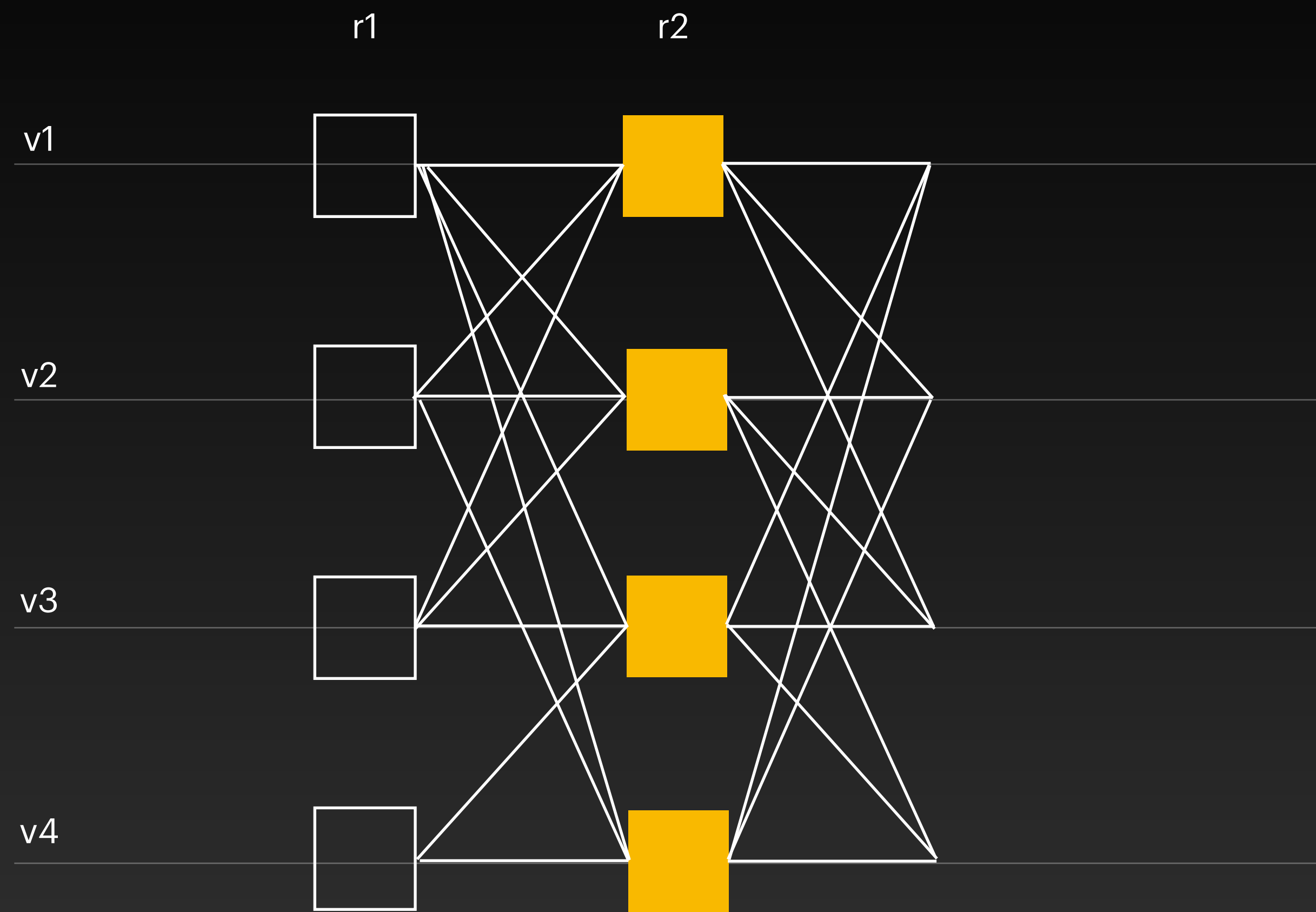


- Round number
- Author
- Payload (txs)
- Parent
- Signature



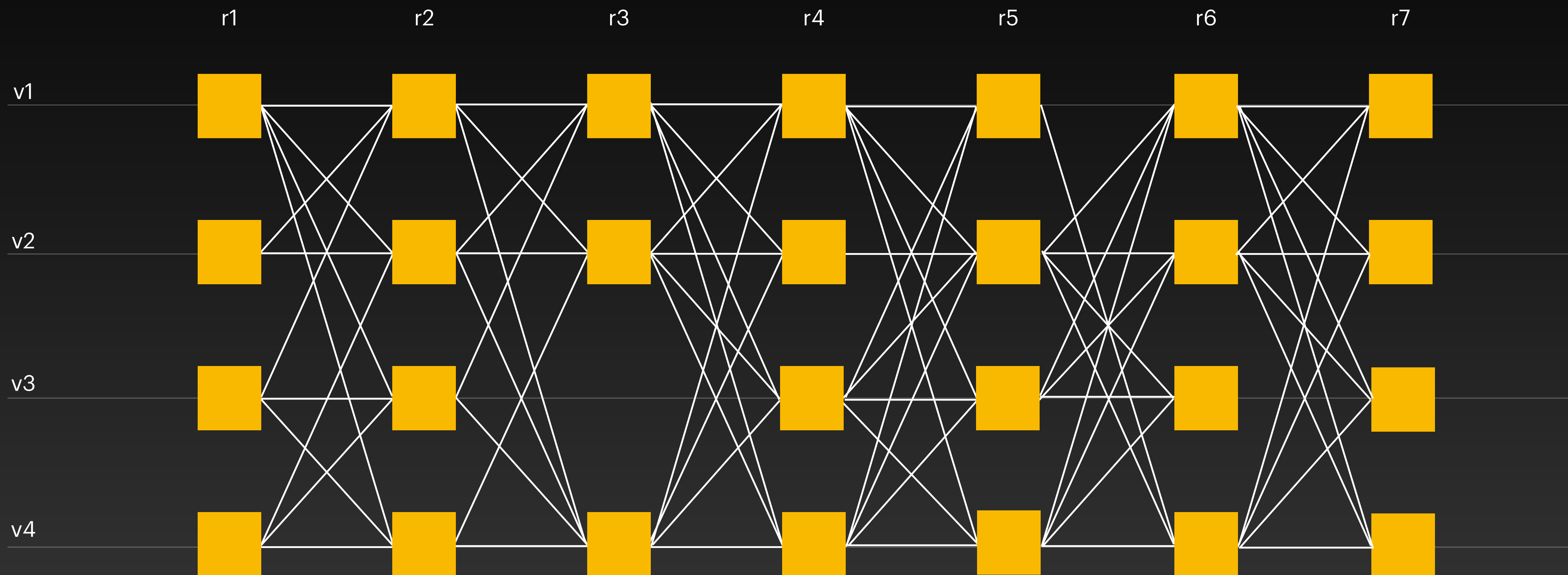
- Round number
- Author
- Payload (txs)
- **Parents**
- Signature

Back Pressure



- Round number
- Author
- Payload (txs)
- Parents
- Signature

High Resource Utilisation

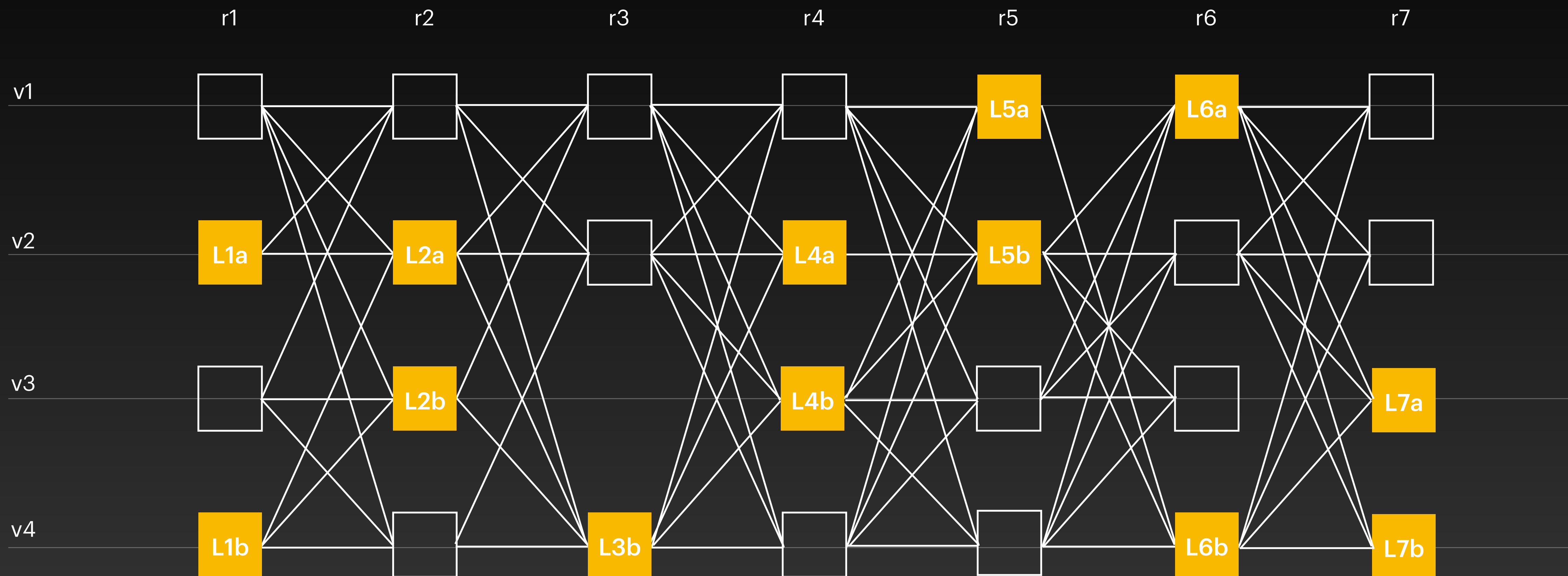


Back Pressure

Robustness

High Resource Utilisation

Throughput



Back Pressure

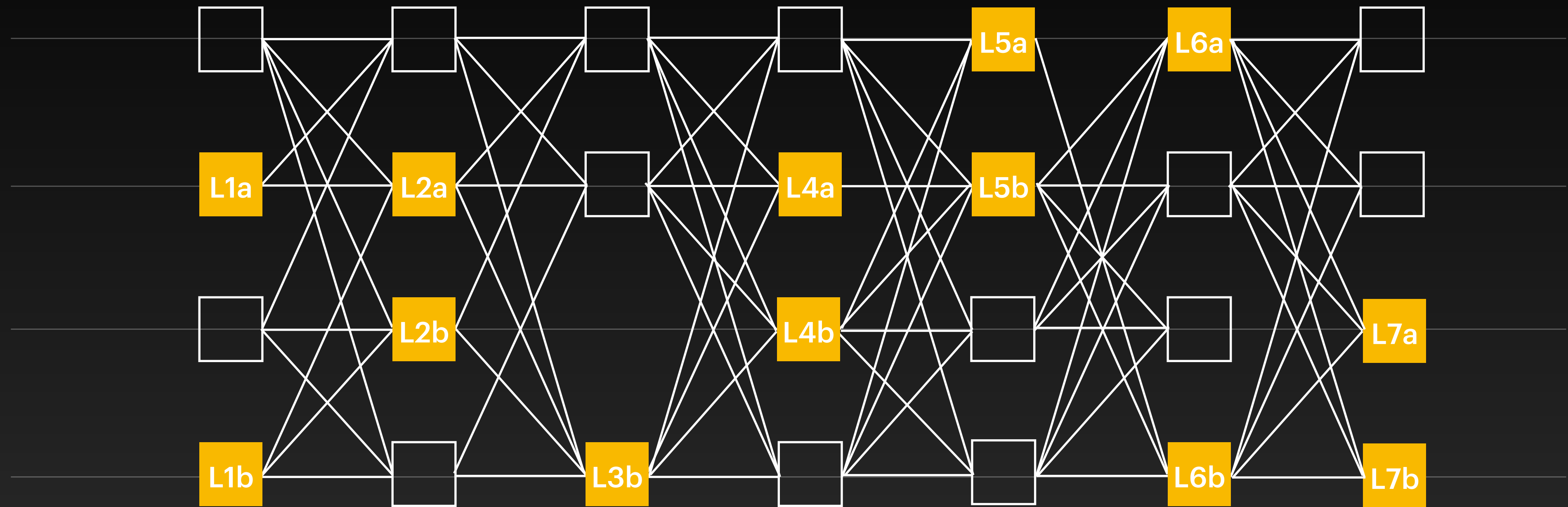
Robustness

High Resource Utilisation

Throughput

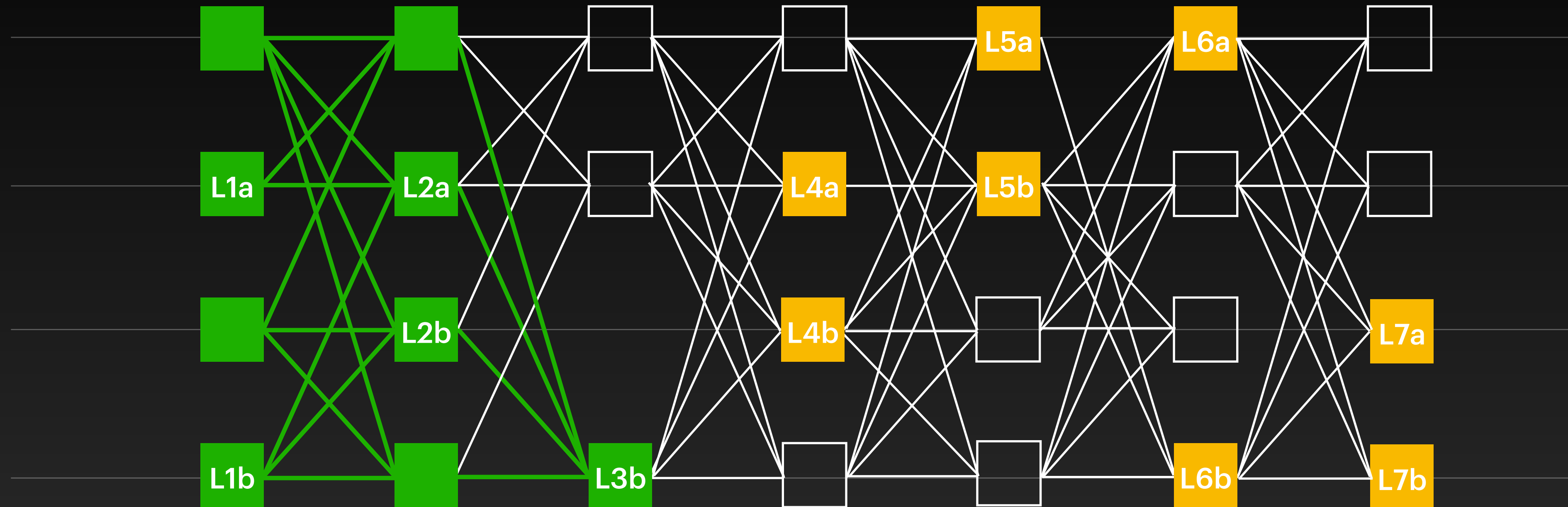
Frequent Commits

Latency



The goal is to order leaders:

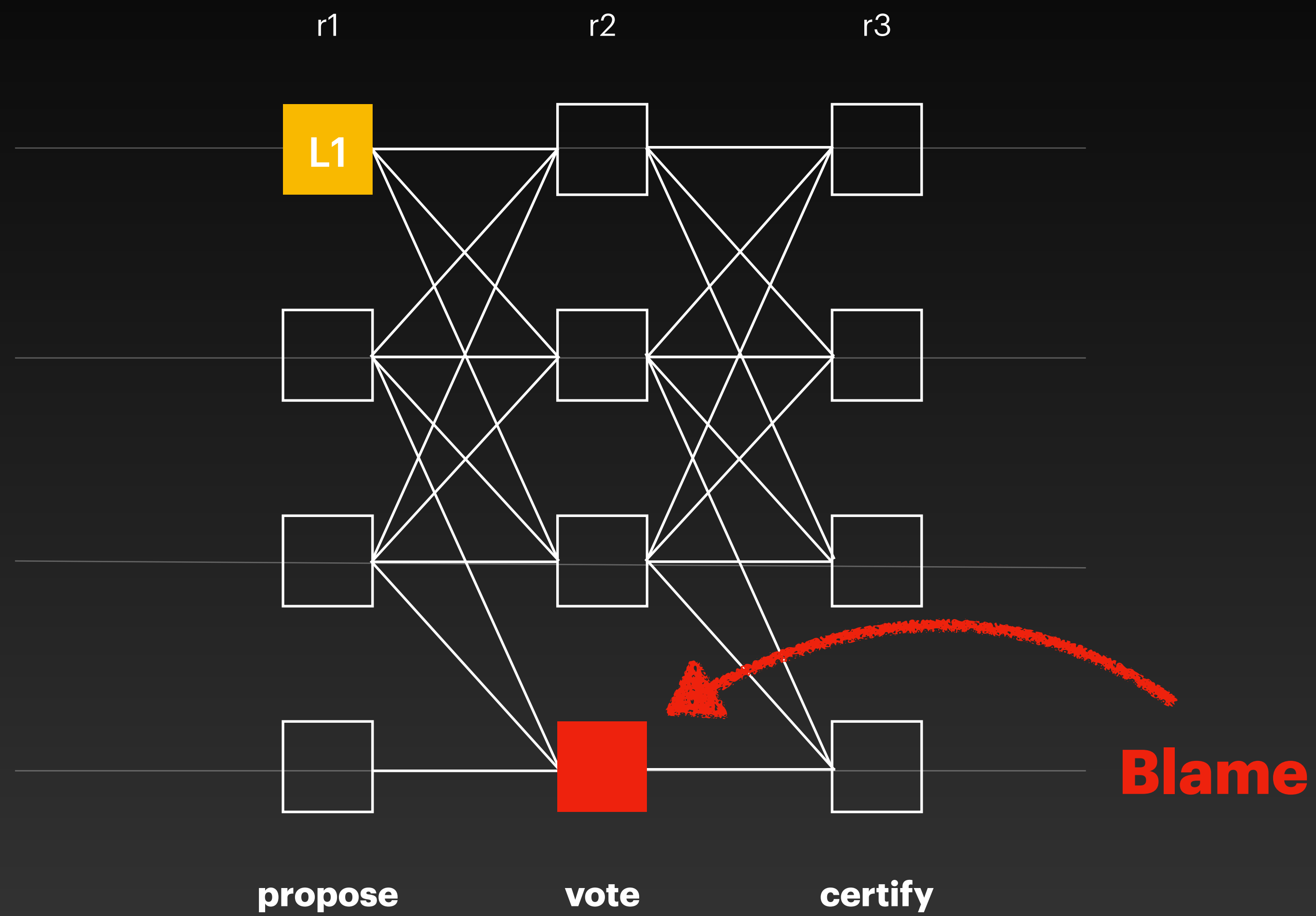


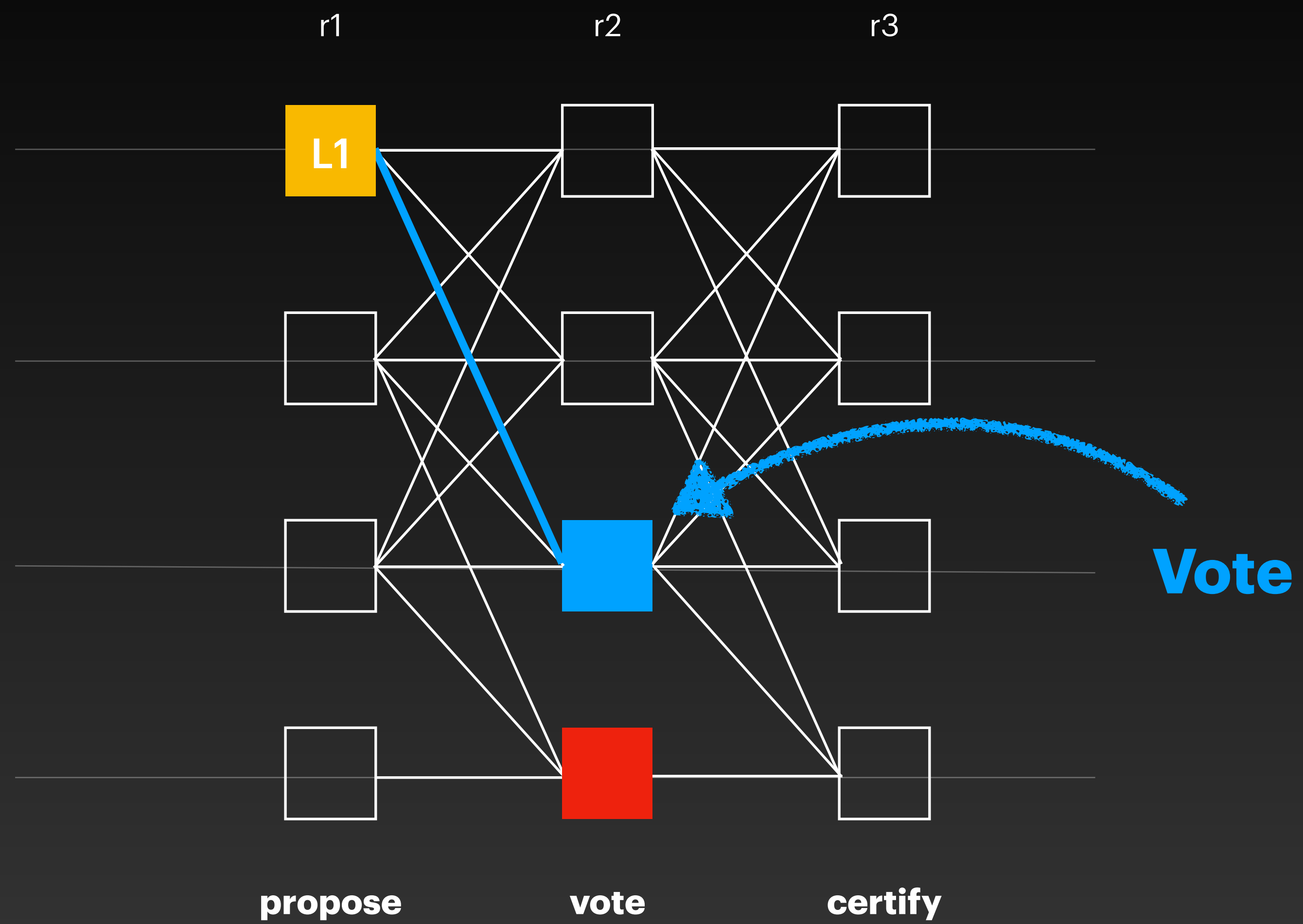


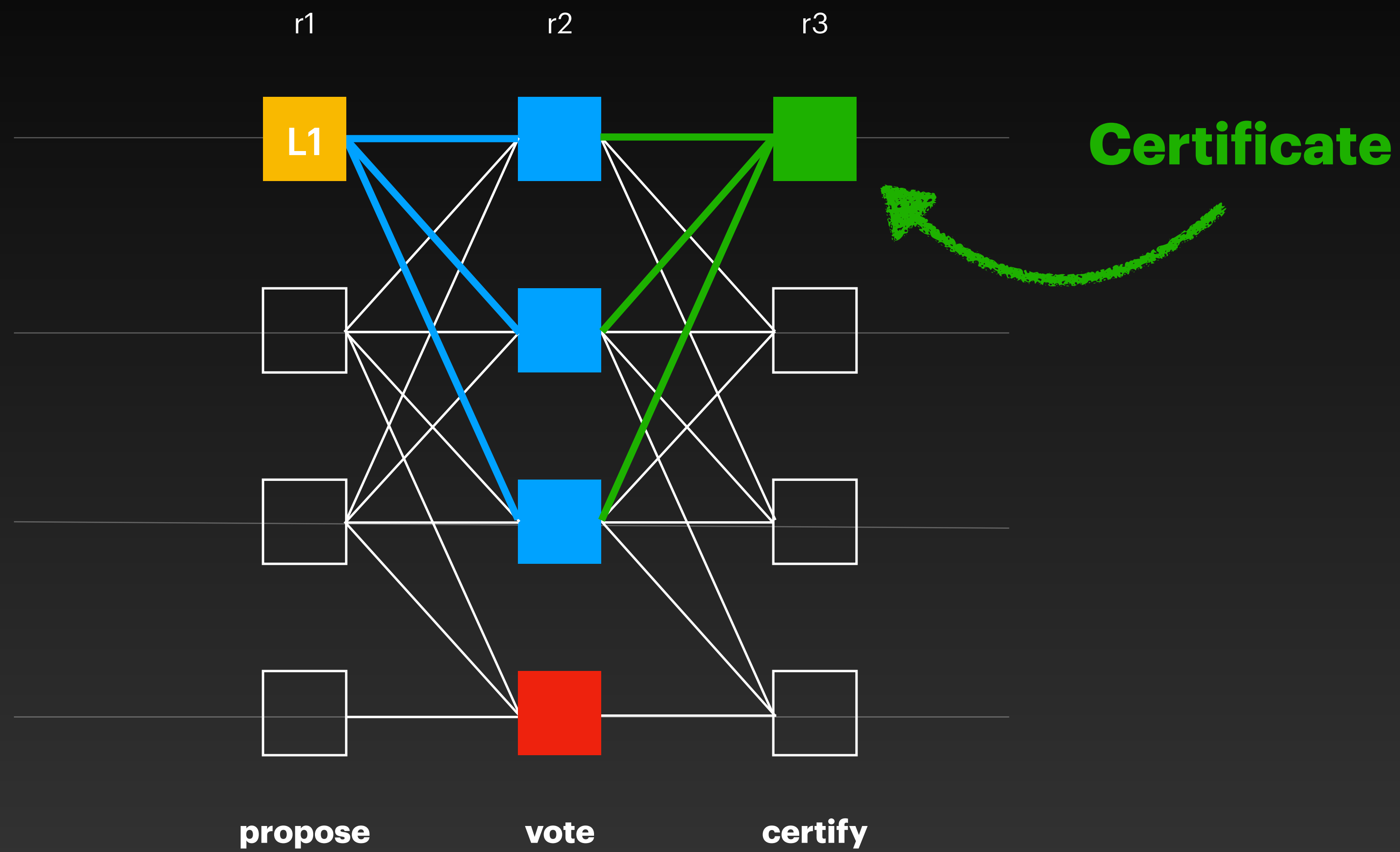
The goal is to order leaders:

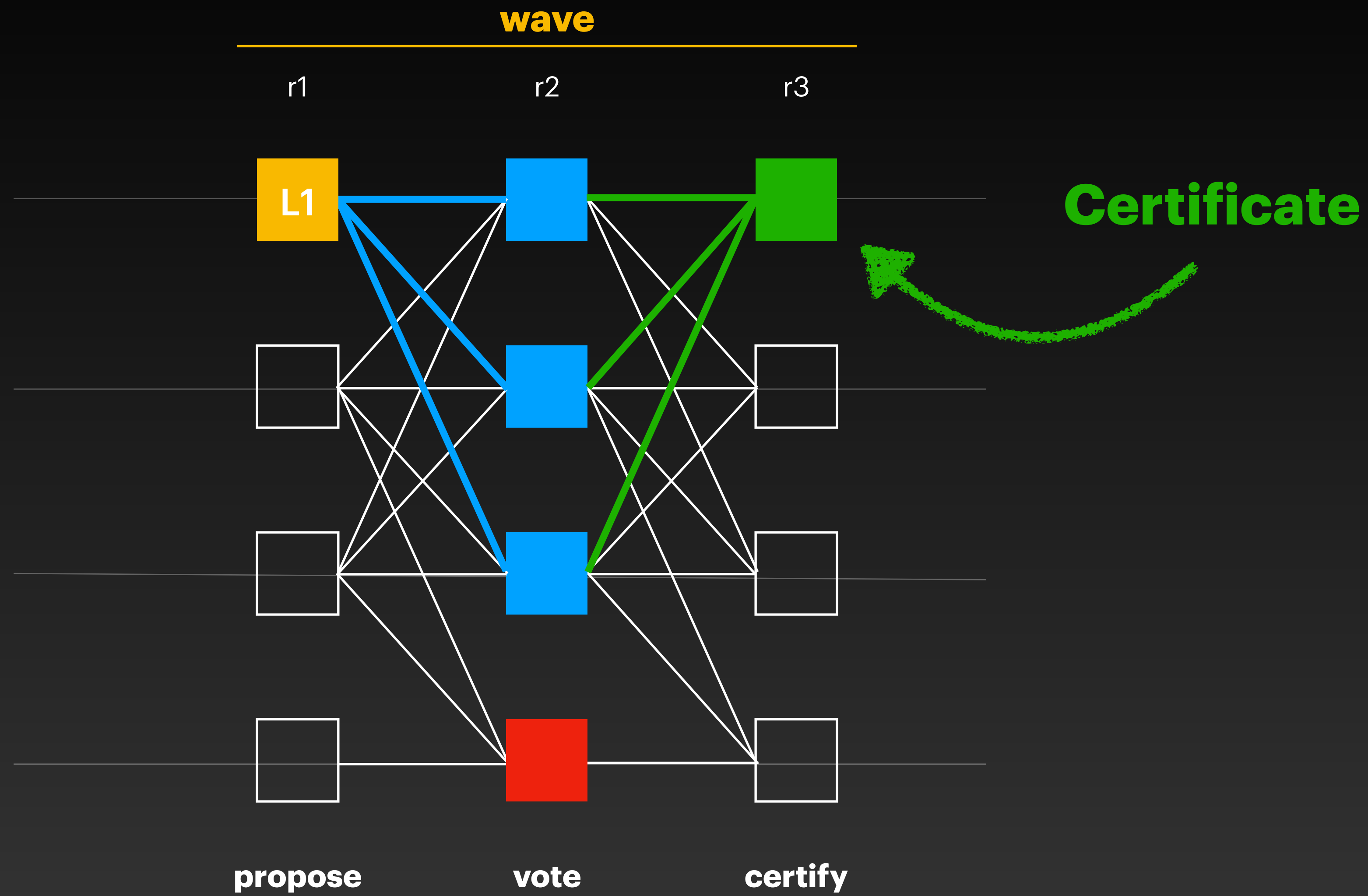


Linearising the sub-dag is simple





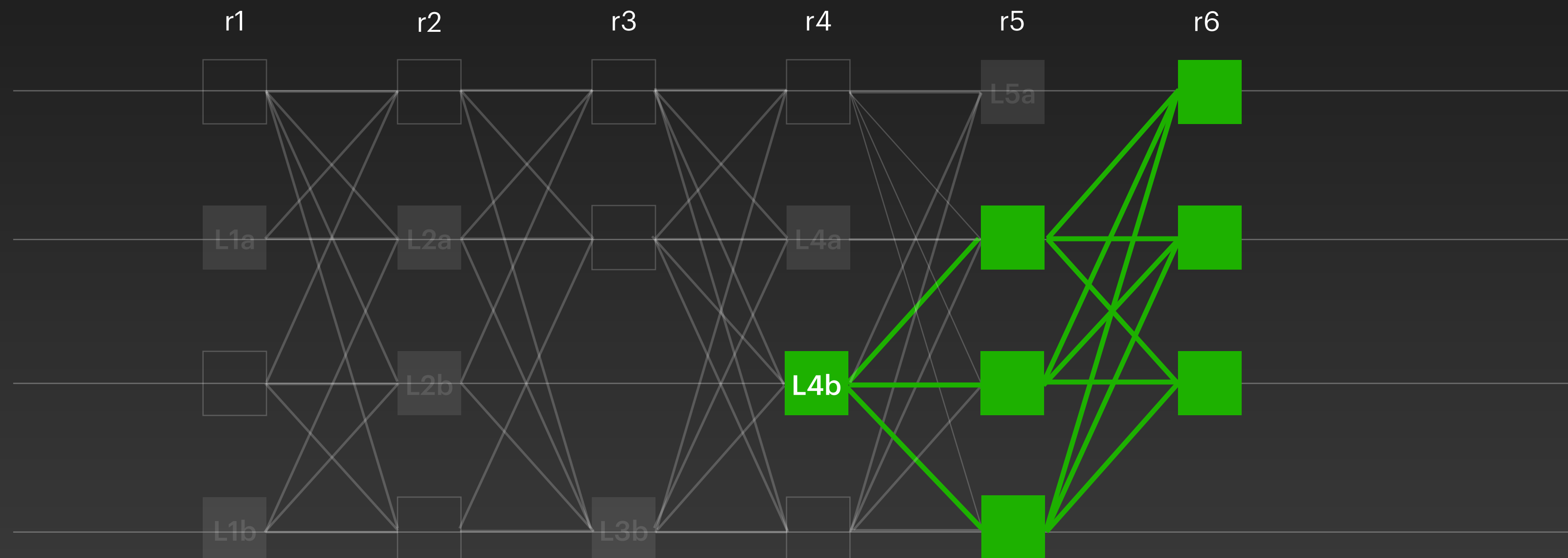




Direct decision rule

On each leader starting from highest round

- **Skip** if enough **blames**
- **Commit** if enough **votes** or **certificates**
- **Undecided** otherwise

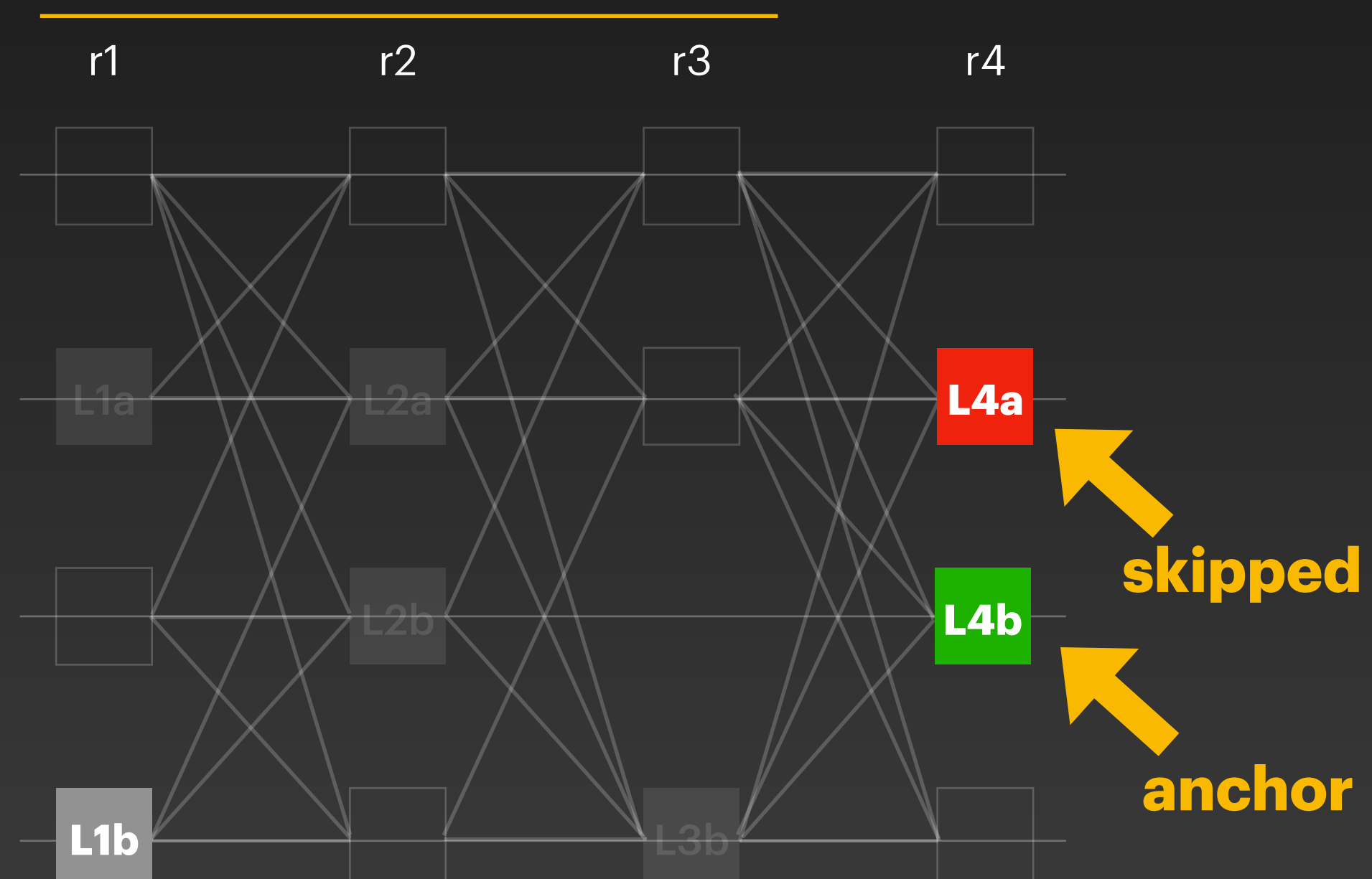


Indirect decision rule

On each undecided leader starting from the highest round

(i) find the anchor:

- First block with round $> r+2$ that is **Commit** or **Undecided**

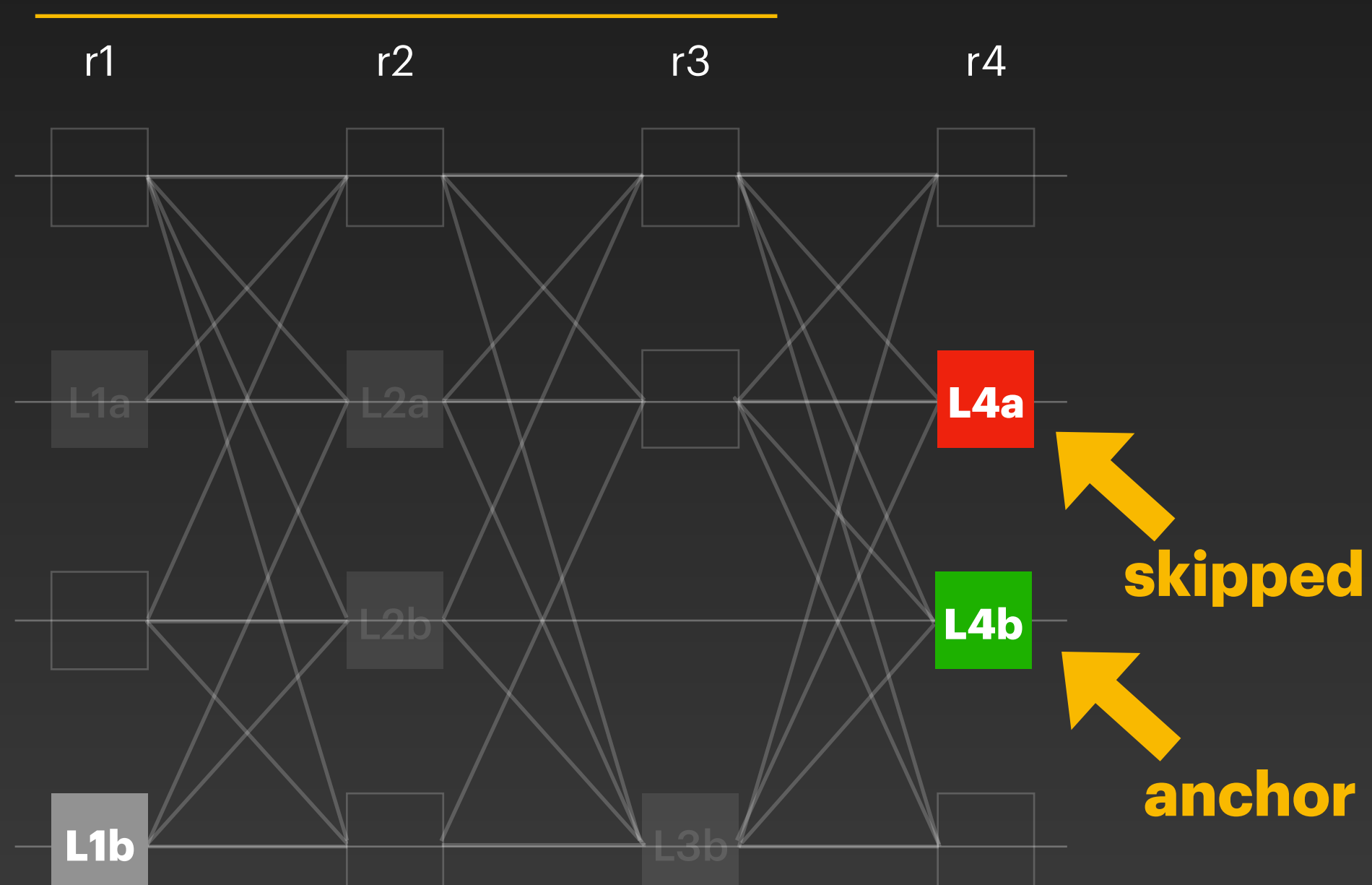


Indirect decision rule

On each undecided leader starting from the highest round

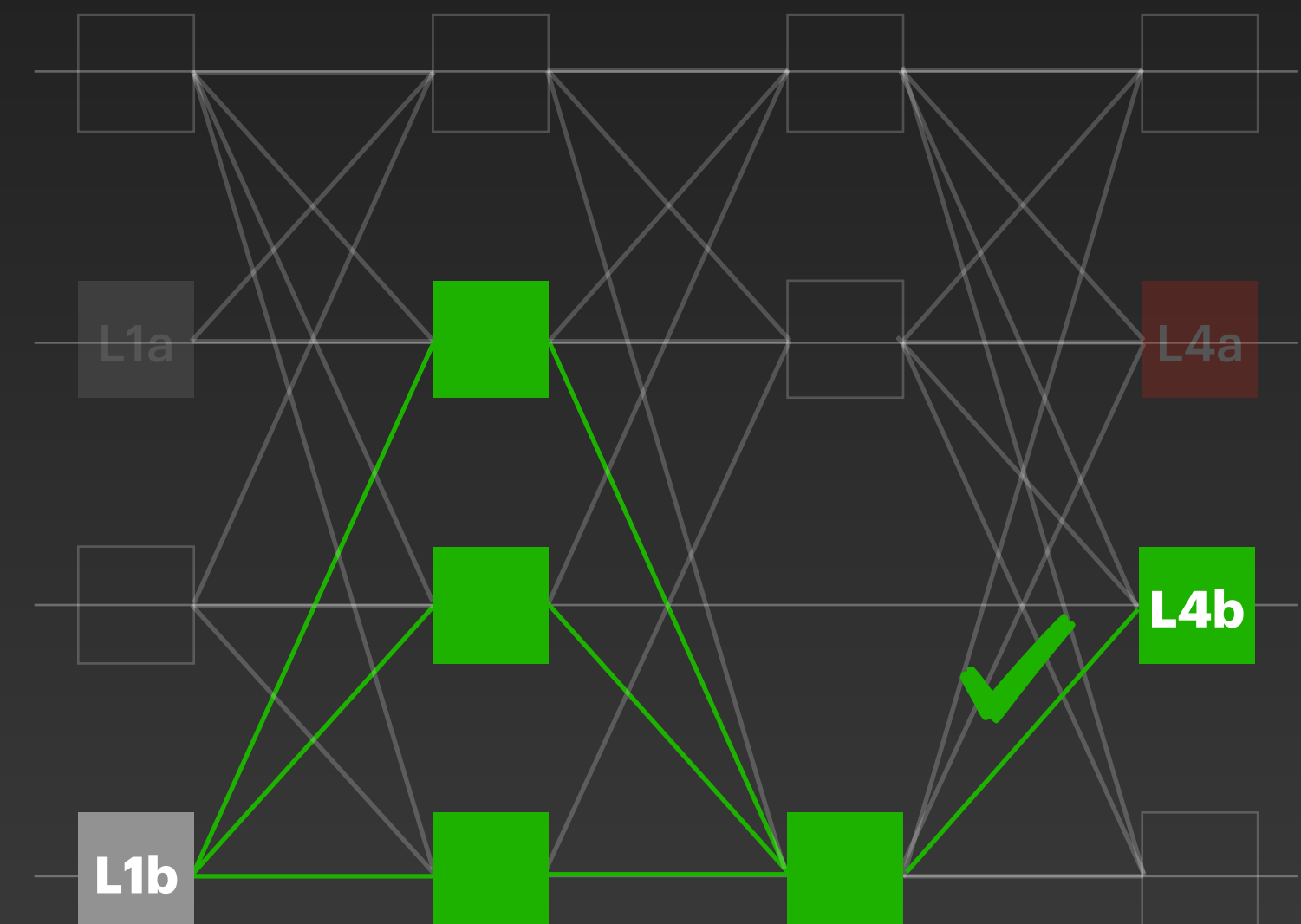
(i) find the anchor:

- First block with round $> r+2$ that is **Commit** or **Undecided**



(ii) check anchor link:

- Commit** if enough **votes** or **certificates**
- Skip** otherwise



$$n \geq 3f + c + 2p + 1$$

$$n \geq 3f + c + 2p + 1$$

Byzantine

Crashed

Fast-Path

$$\text{parents} \geq n - f - c$$

$$n \geq 3f + c + 2p + 1$$

Byzantine

Crashed

Fast-Path

$$n \geq 3f + \cancel{c} + \cancel{2p} + 1$$

Mysticeti

$$n \geq 3f + \cancel{c} + \cancel{2p} + 1$$

Mysticeti

$$n \geq 3f + \cancel{c} + 2\mathbf{f} + 1$$

BlueBottle

$$n \geq 3f + \cancel{c} + \cancel{2p} + 1$$

Mysticeti

$$n \geq 3f + \cancel{c} + 2\mathbf{f} + 1$$

BlueBottle

$$n \geq 3f + c + 2(\mathbf{f+c}) + 1$$

Orcaella

$$n \geq 3f + \cancel{c} + \cancel{2p} + 1$$

Mysticeti

$$n \geq 3f + \cancel{c} + 2f + 1$$

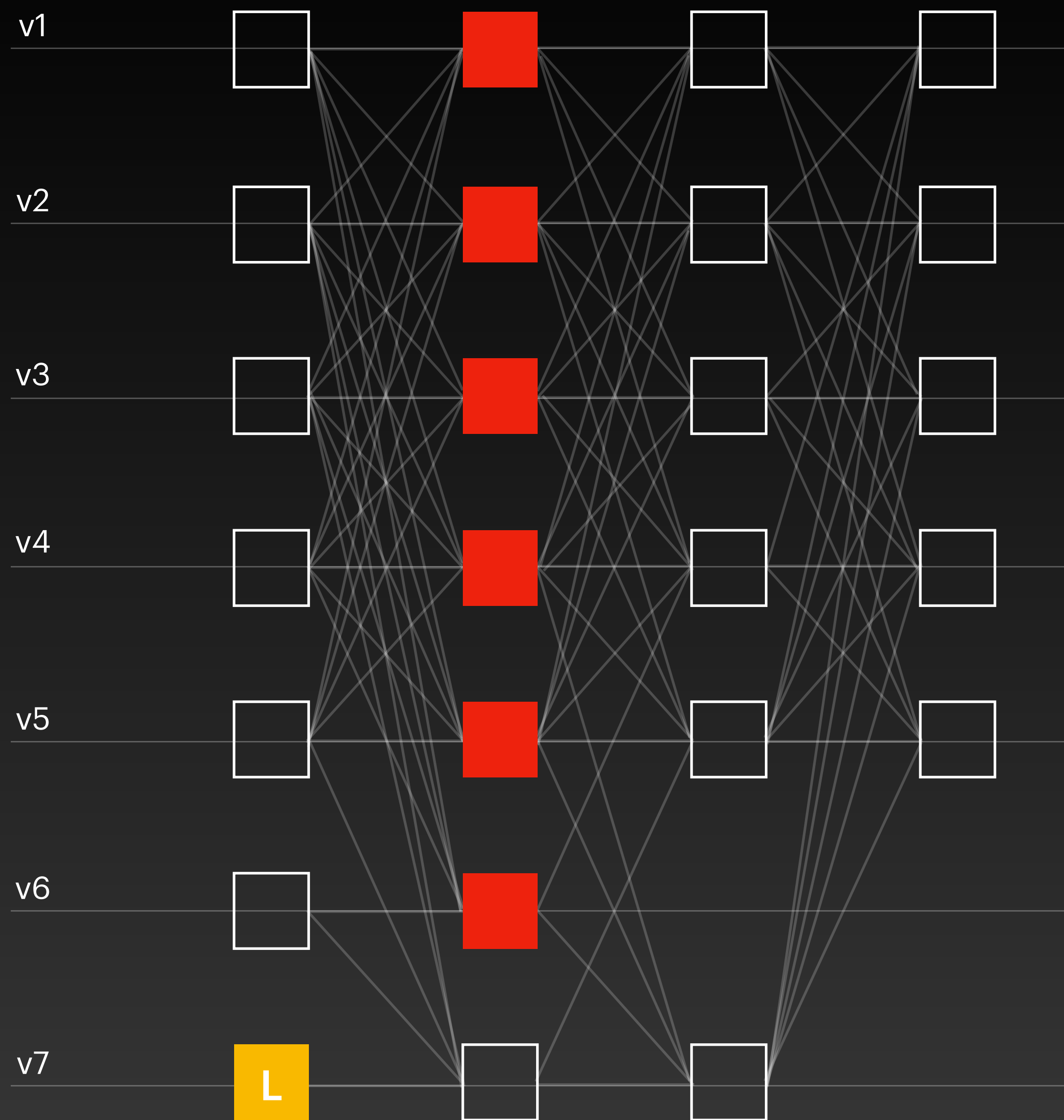
BlueBottle

$$n \geq 3f + c + 2(f+c) + 1$$

Orcaella

$$n \geq \cancel{3f} + c + 2(c/2) + 1$$

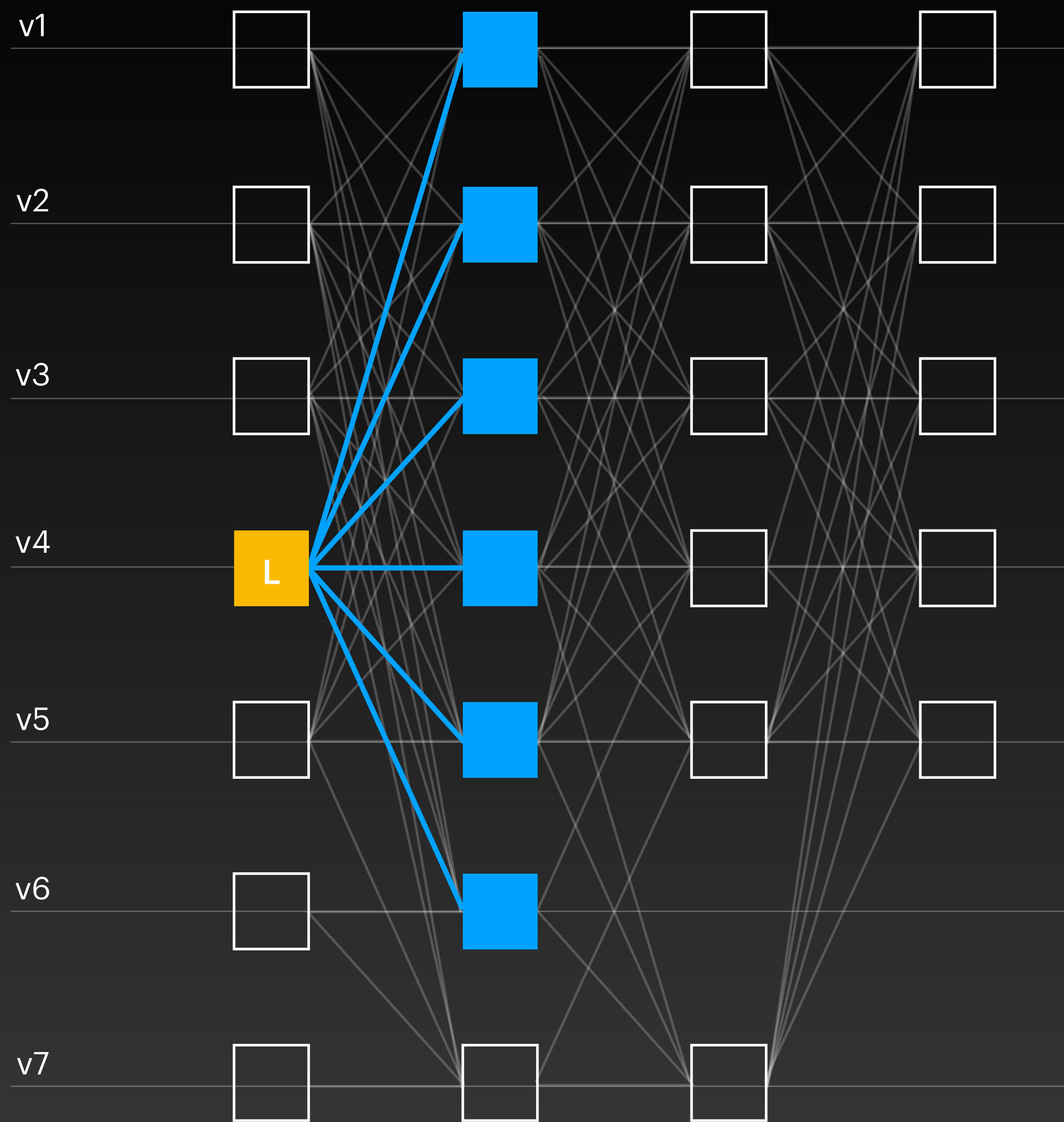
NemoNemo



Direct decision rule

$$f = c = p = 1$$

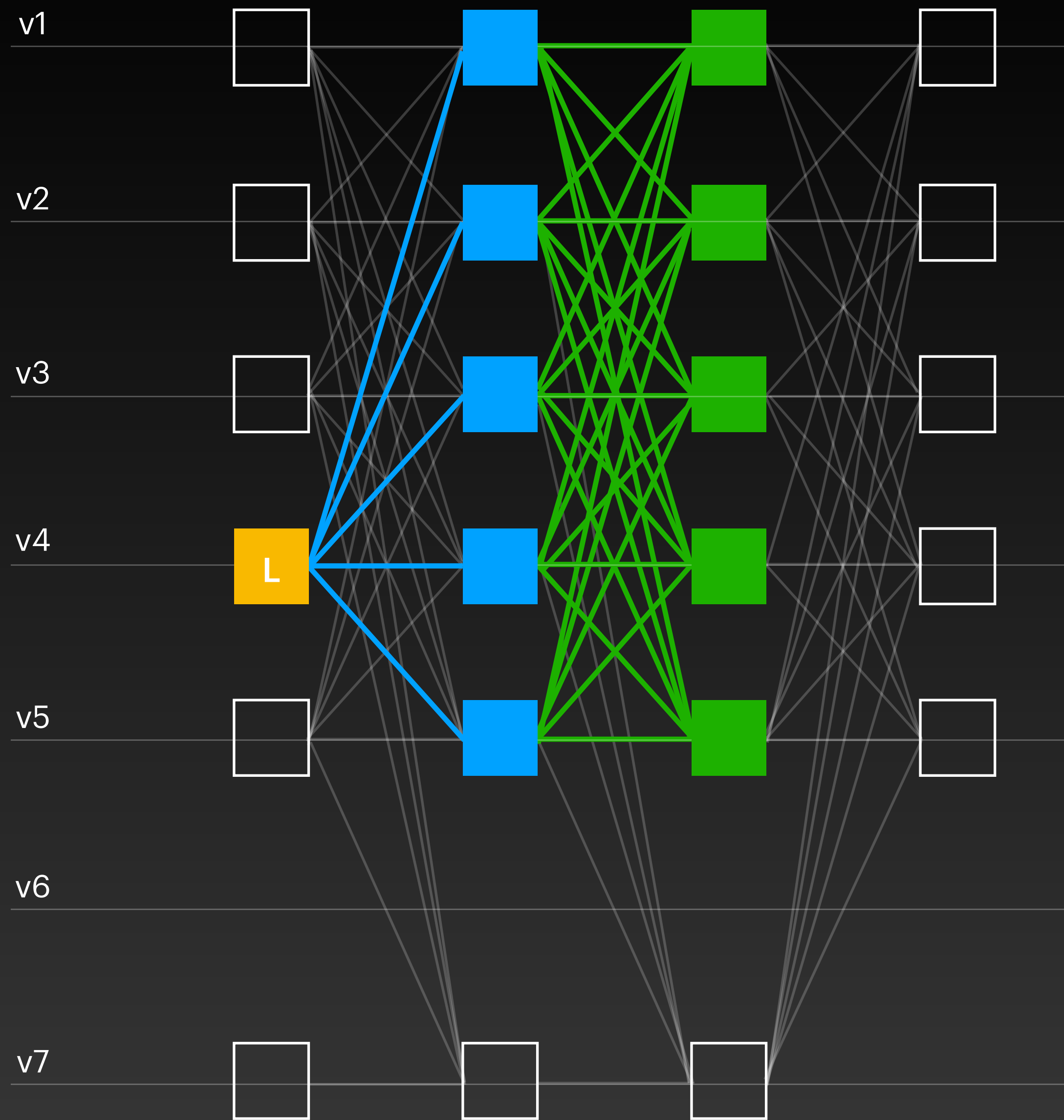
- **Skip** if $\geq n-p$ blames



Direct decision rule

$$f = c = p = 1$$

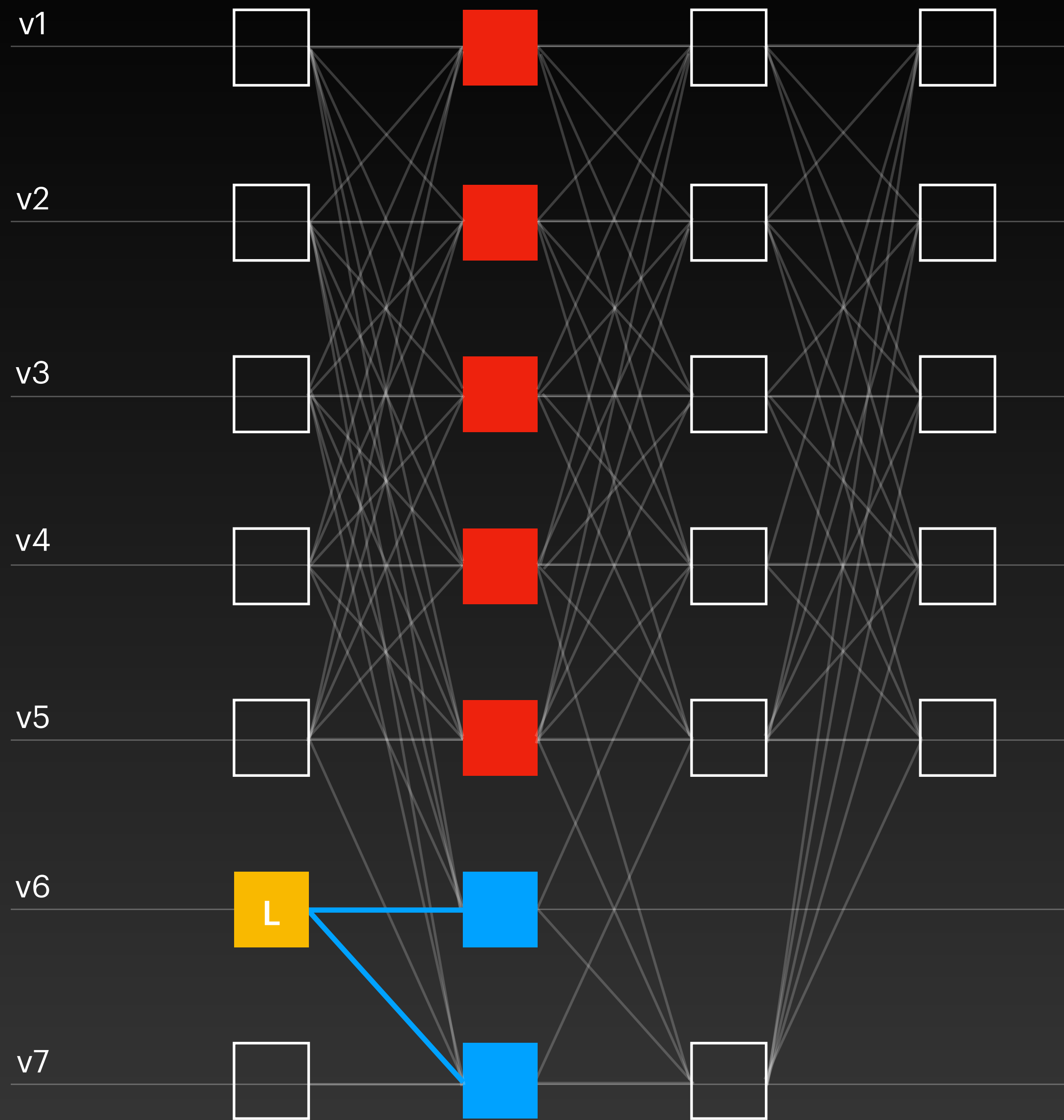
- **Skip** if $\geq n-p$ blames
- **Commit** if $\geq n-p$ votes



Direct decision rule

$$f = c = p = 1$$

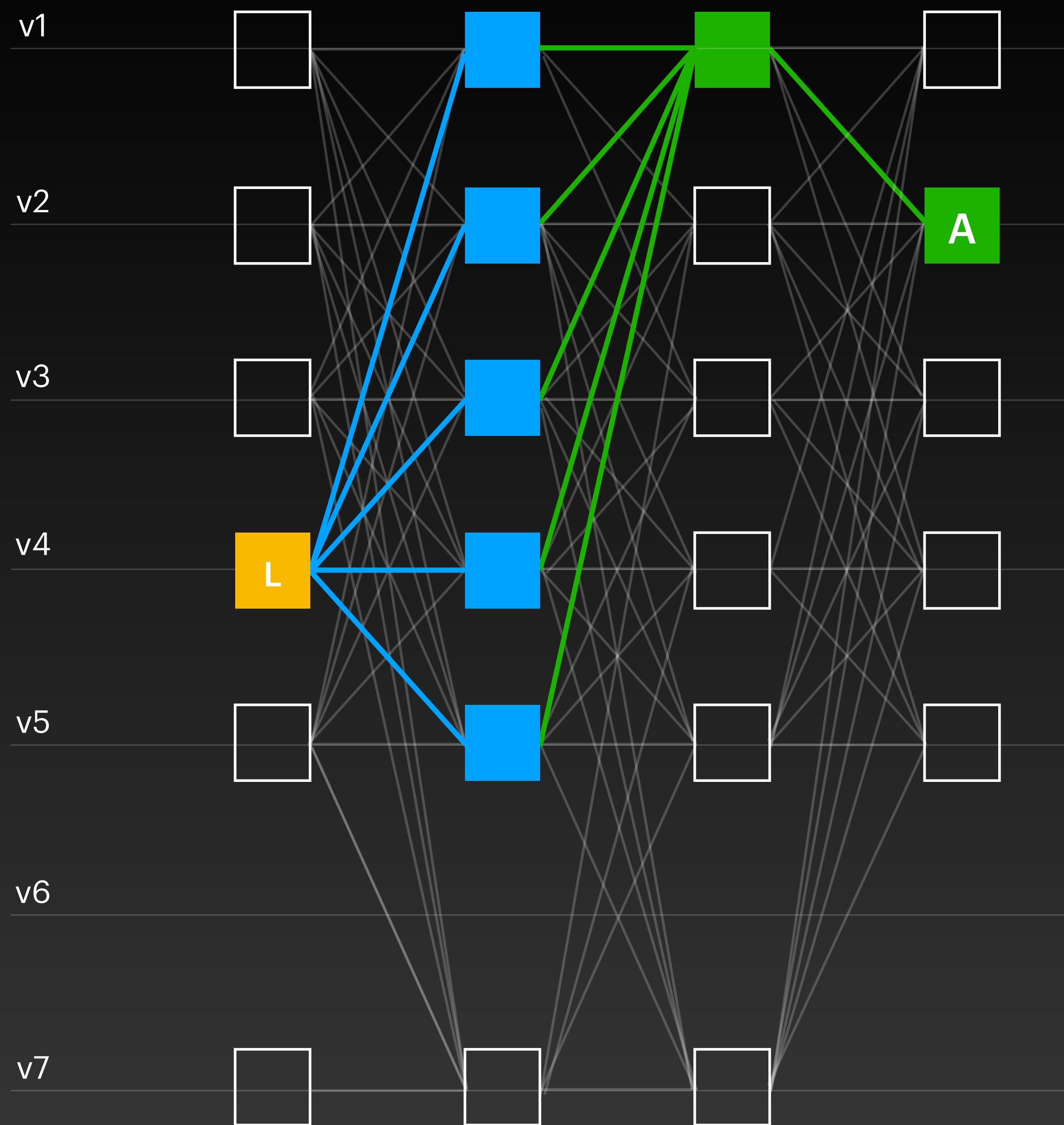
- **Skip** if $\geq n-p$ **blames**
- **Commit** if $\geq n-p$ **votes**
- **Commit** if $\geq 2f+c+1$ **certificates** made of $\geq (n+f+1)/2$ **votes**



Direct decision rule

$$f = c = p = 1$$

- **Skip** if $\geq n-p$ **blames**
- **Commit** if $\geq n-p$ **votes**
- **Commit** if $\geq 2f+c+1$ **certificates** made of $\geq (n+f+1)/2$ **votes**
- **Undecided** otherwise

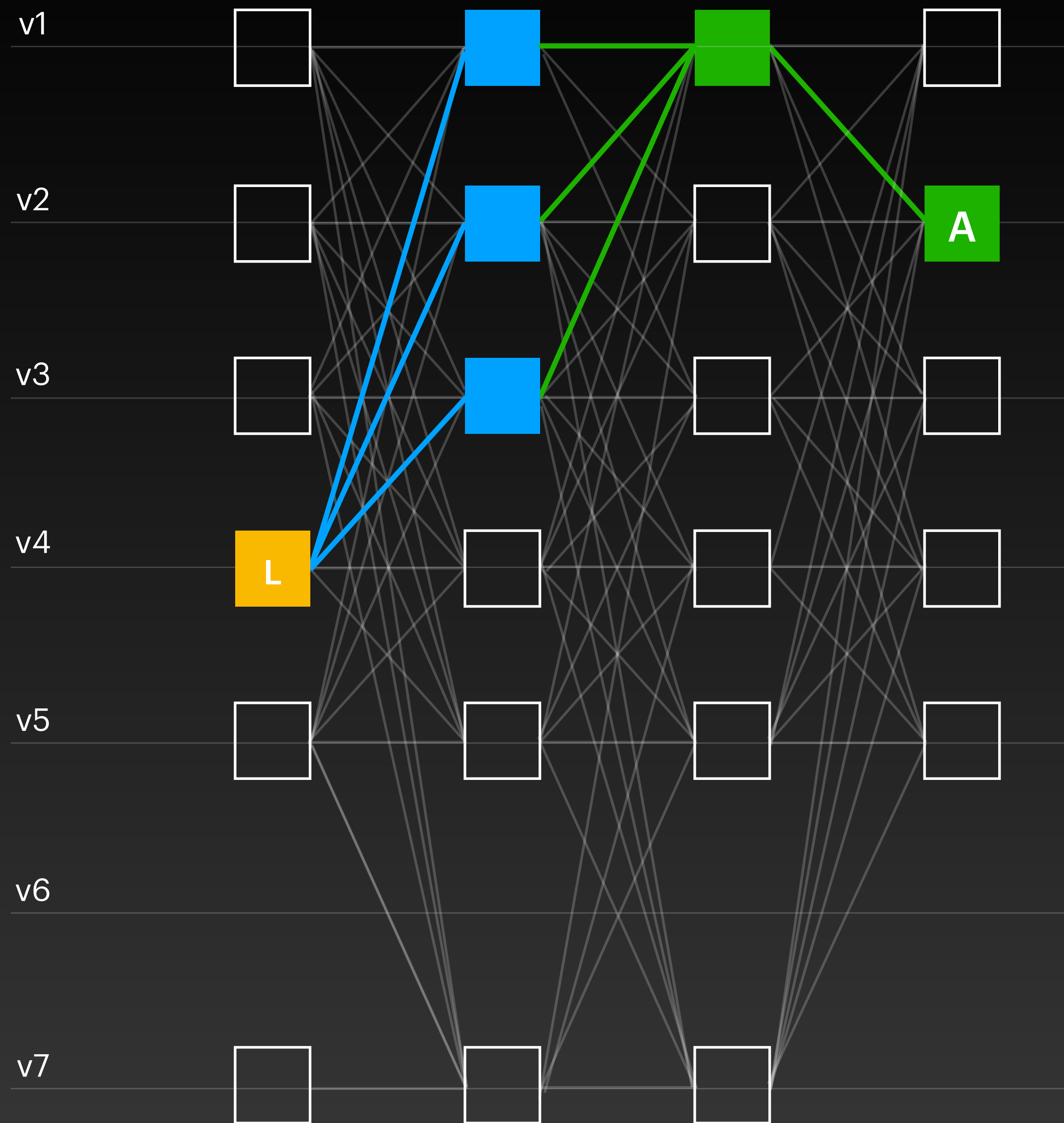


Indirect decision rule

$$f = c = p = 1$$

Anchor links to:

- (i) **Certificate rung:** ≥ 1 certificate
made of $\geq (n+f+1)/2$ votes

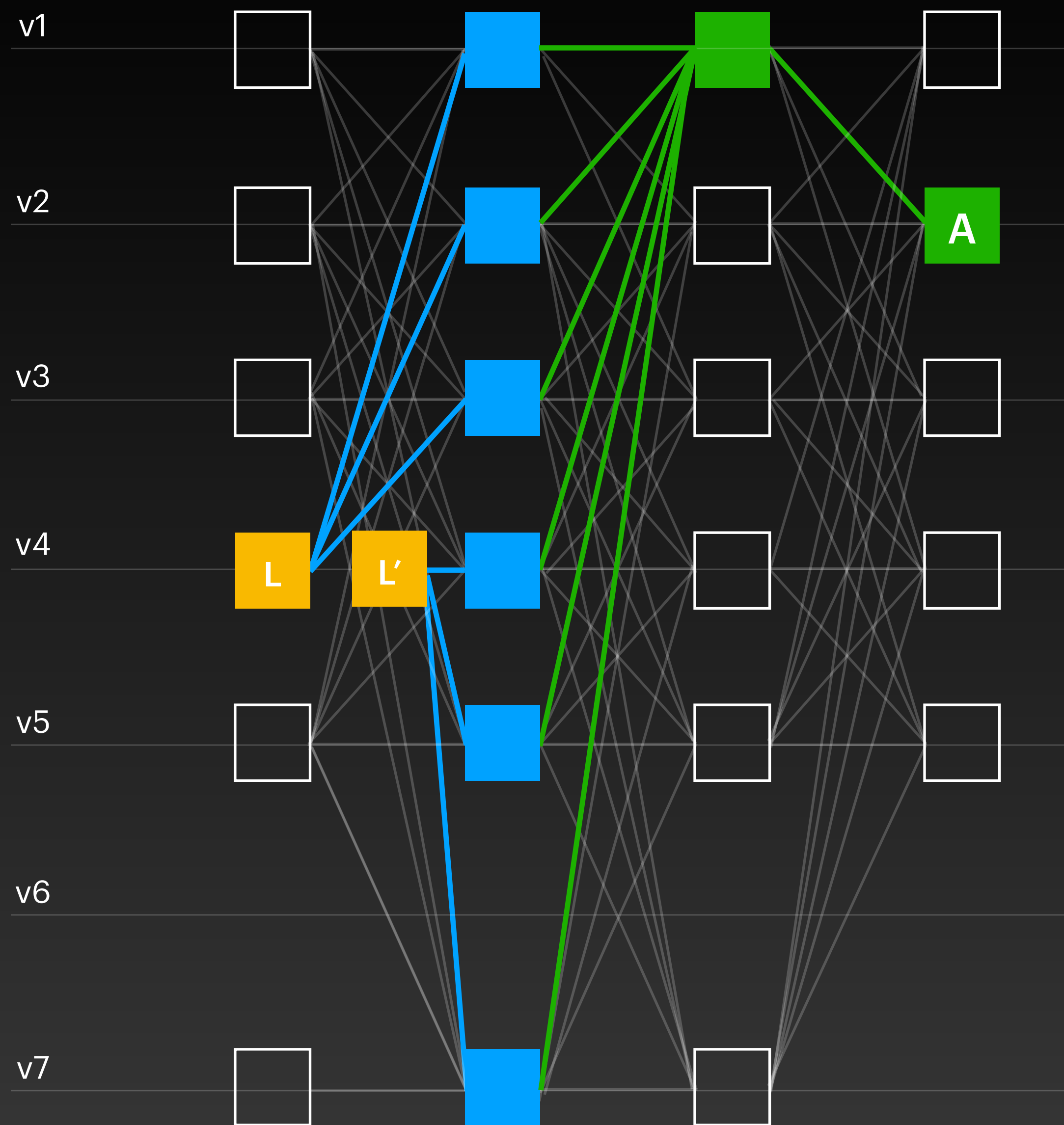


Indirect decision rule

$$f = c = p = 1$$

Anchor links to:

- (i) **Certificate rung:** ≥ 1 certificate made of $\geq (n+f+1)/2$ votes
- (ii) **Weak rung:** ≥ 1 certificate made of $\geq f+p+1$ votes

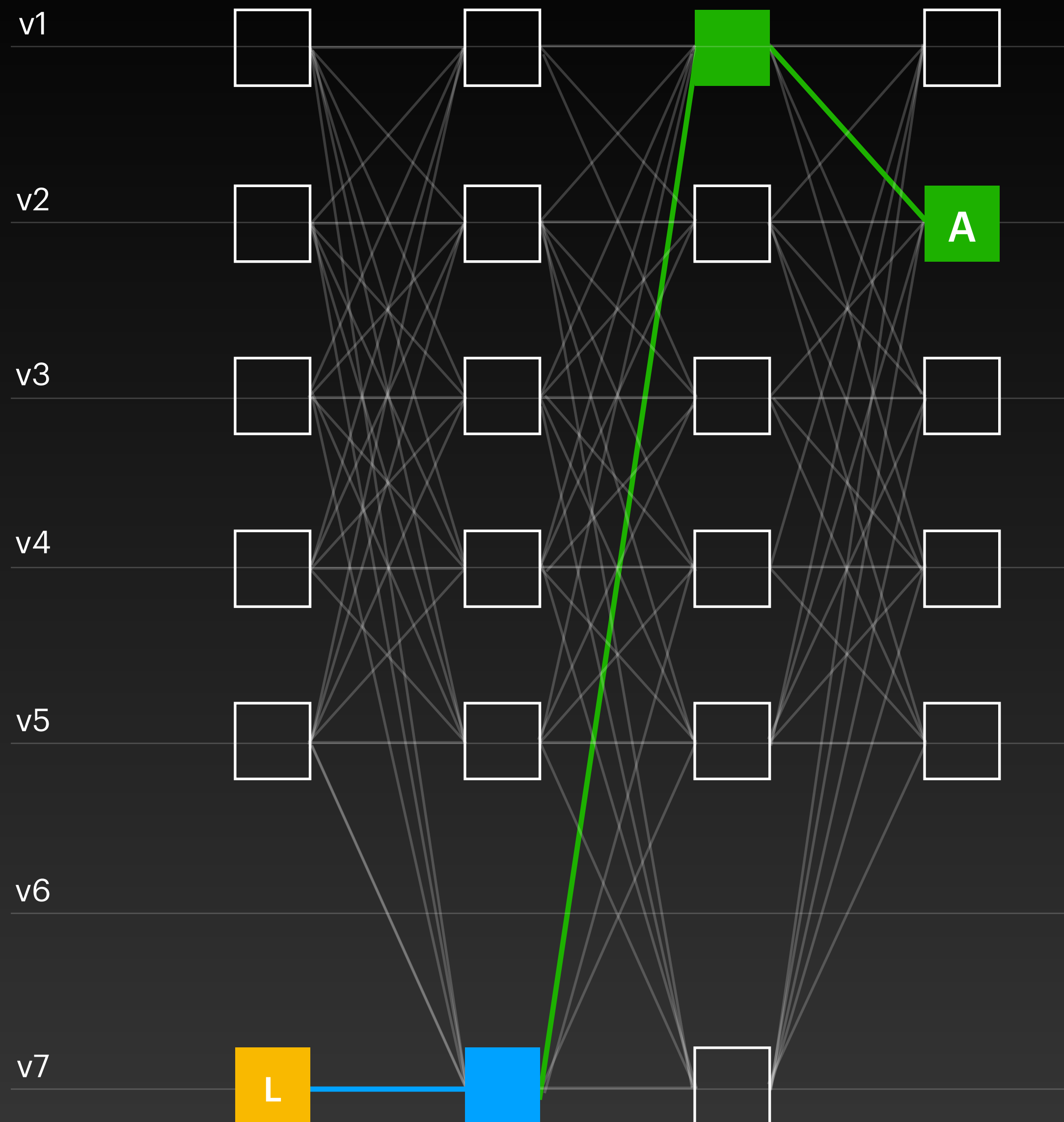


Indirect decision rule

$$f = c = p = 1$$

Anchor links to:

- (i) **Certificate rung:** ≥ 1 certificate made of $\geq (n+f+1)/2$ votes
- (ii) **Weak rung:** ≥ 1 certificate made of $\geq f+p+1$ votes; break ties deterministically



Indirect decision rule

$$f = c = p = 1$$

Anchor links to:

- (i) **Certificate rung:** ≥ 1 certificate made of $\geq (n+f+1)/2$ votes
- (ii) **Weak rung:** ≥ 1 certificate made of $\geq f+p+1$ votes; break ties deterministically
- (iii) **Skip rung:** otherwise

crash faults

24

16

No Fast-Path

Fast-Path

10

16

byzantine faults

$3f+2c+1$

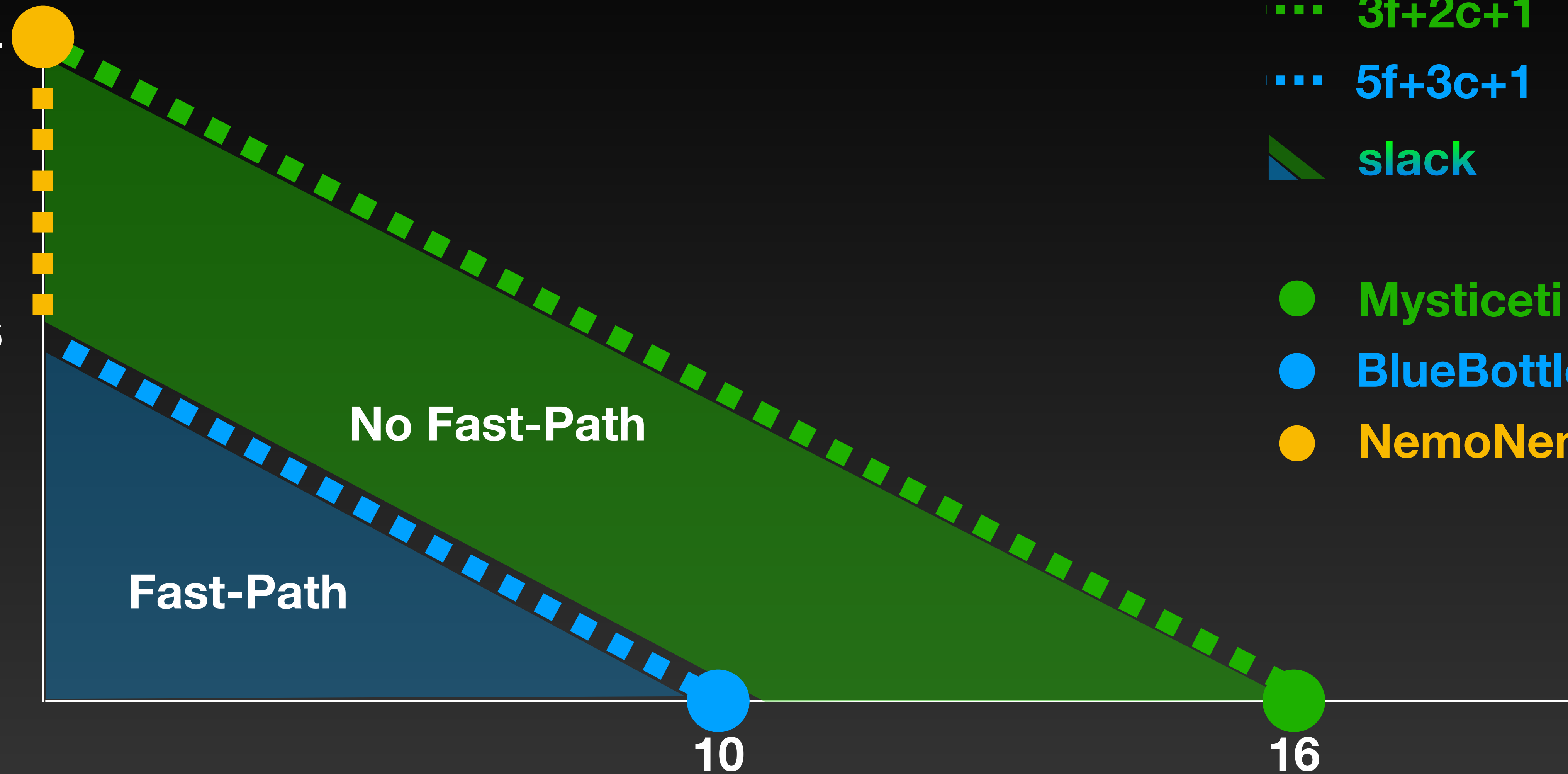
$5f+3c+1$

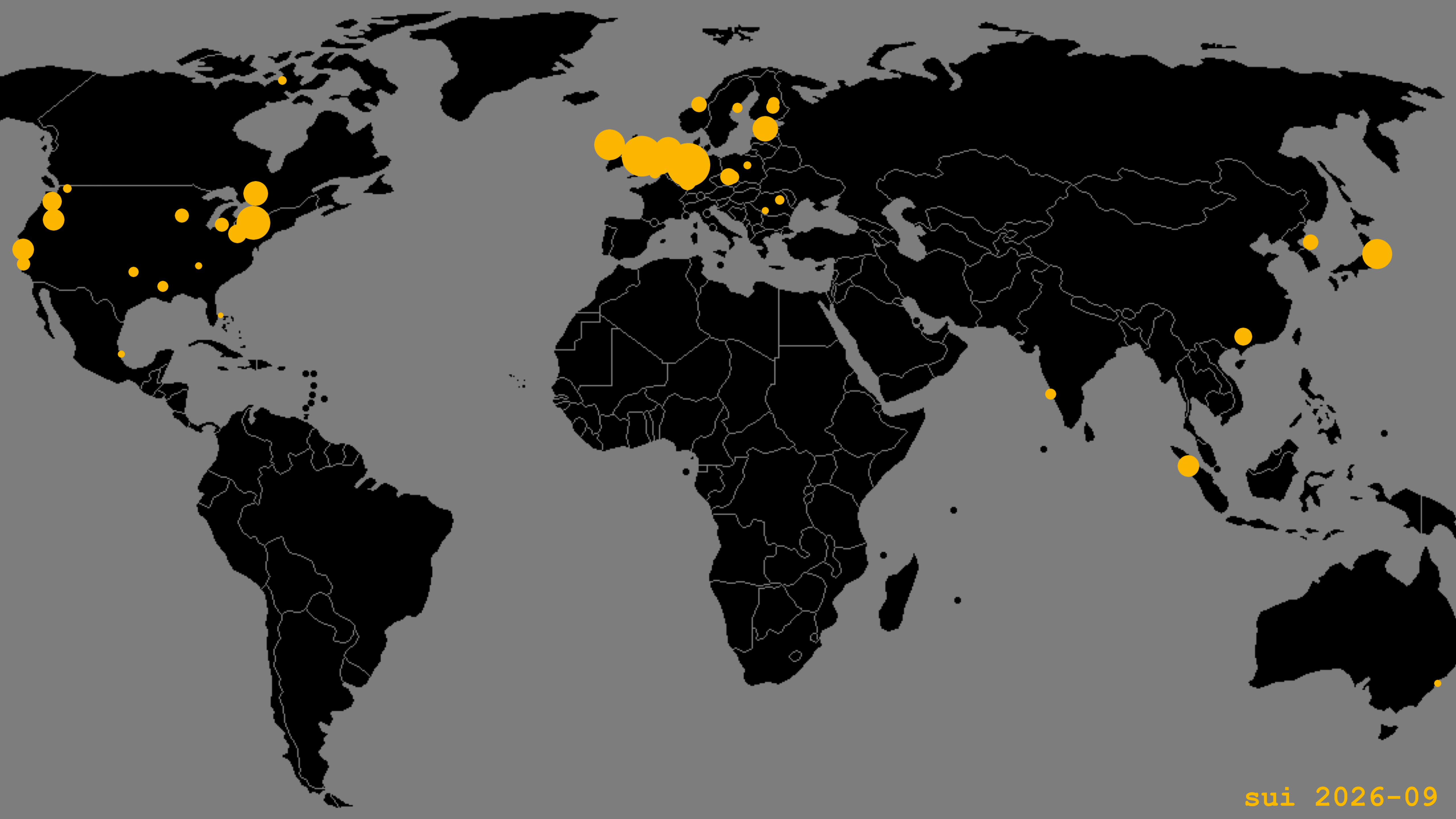
slack

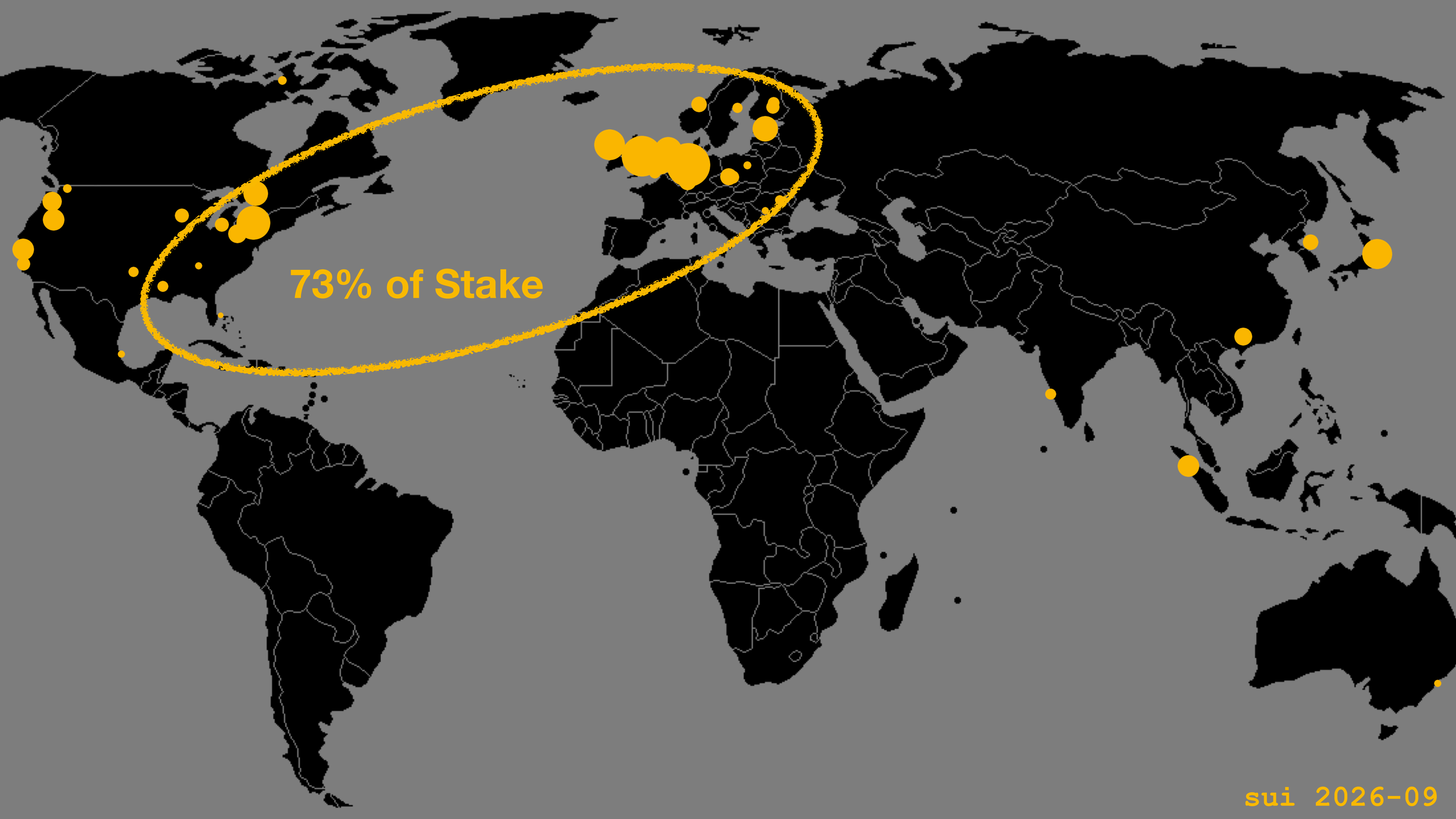
Mysticeti

BlueBottle

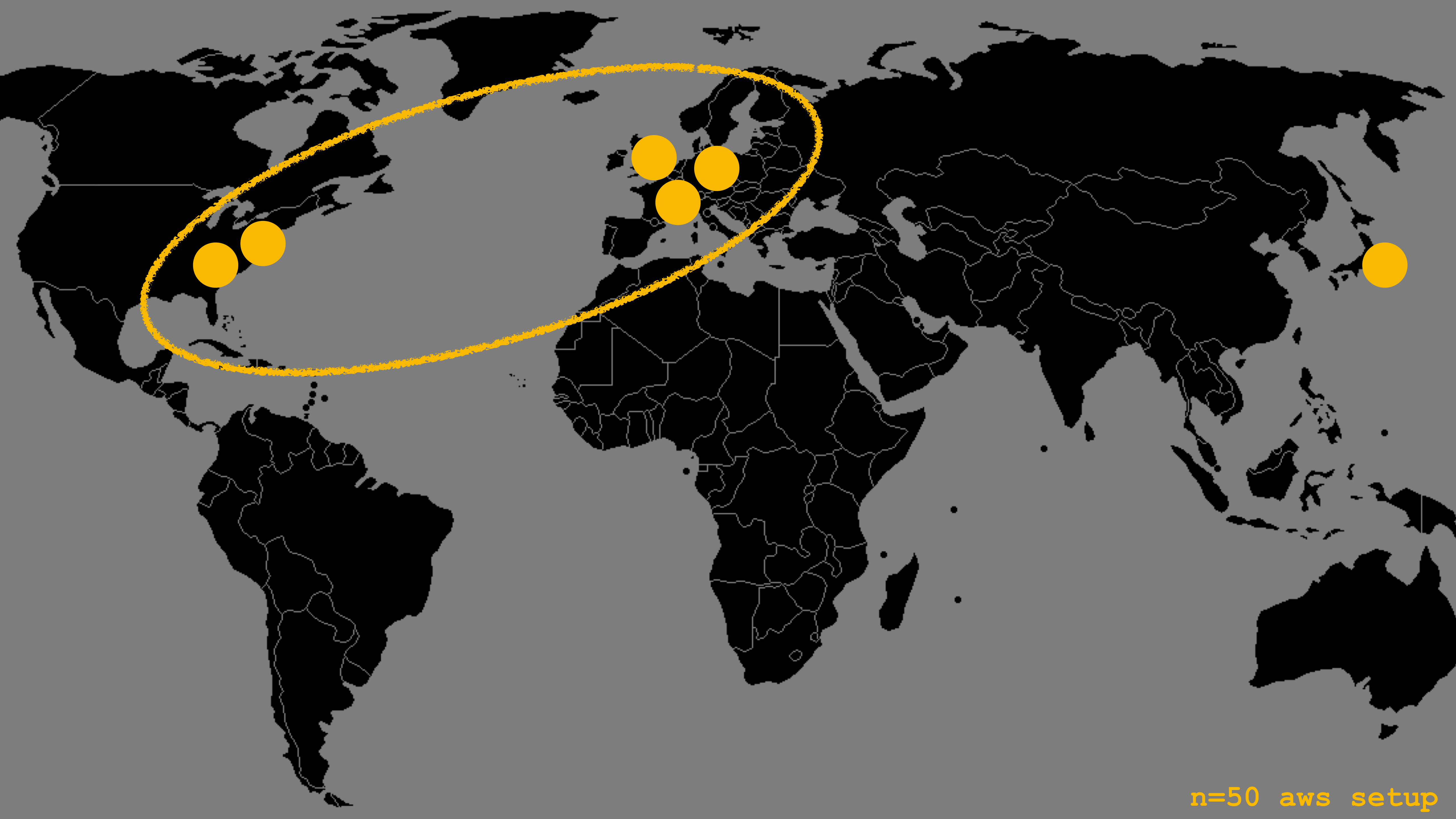
NemoNemo



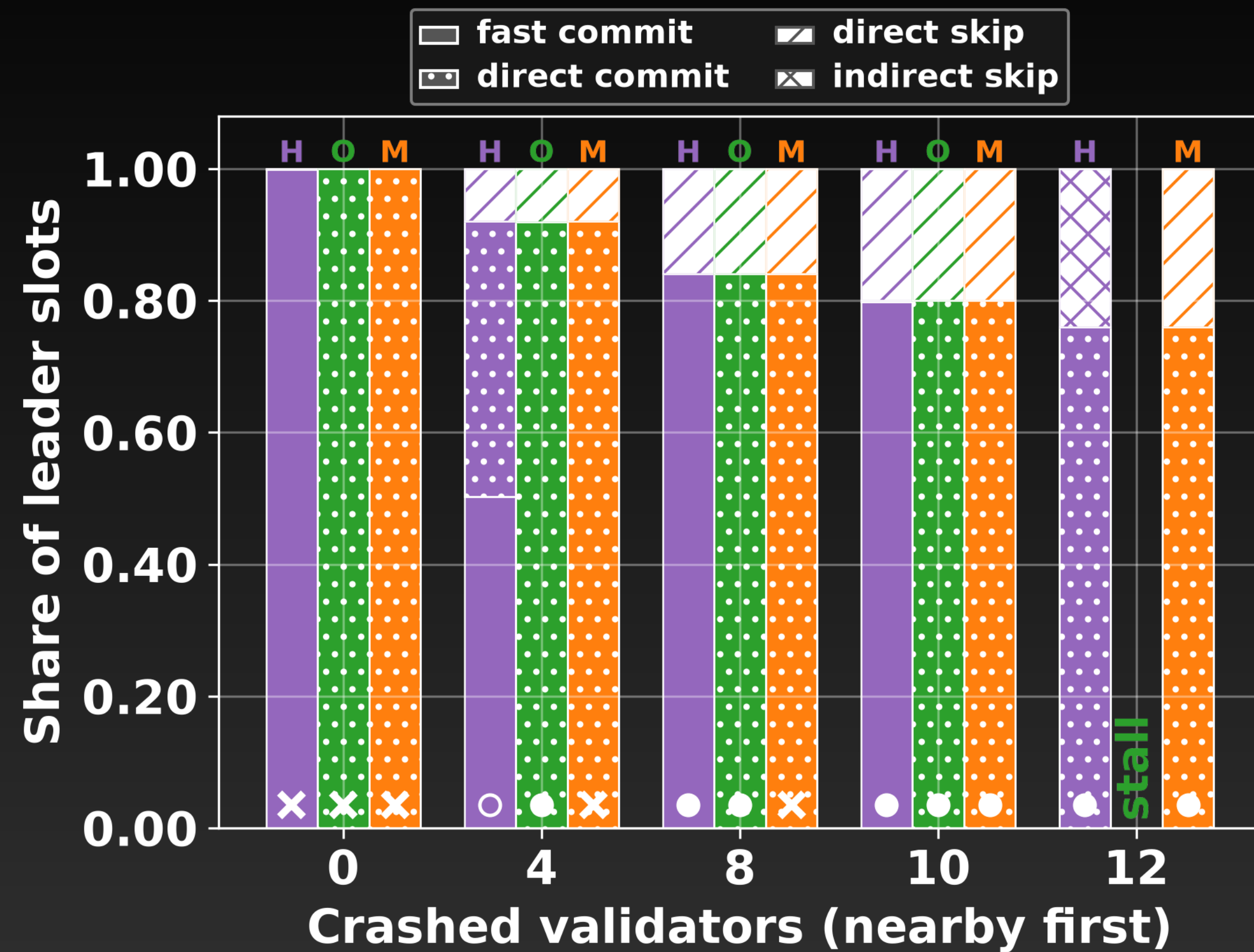
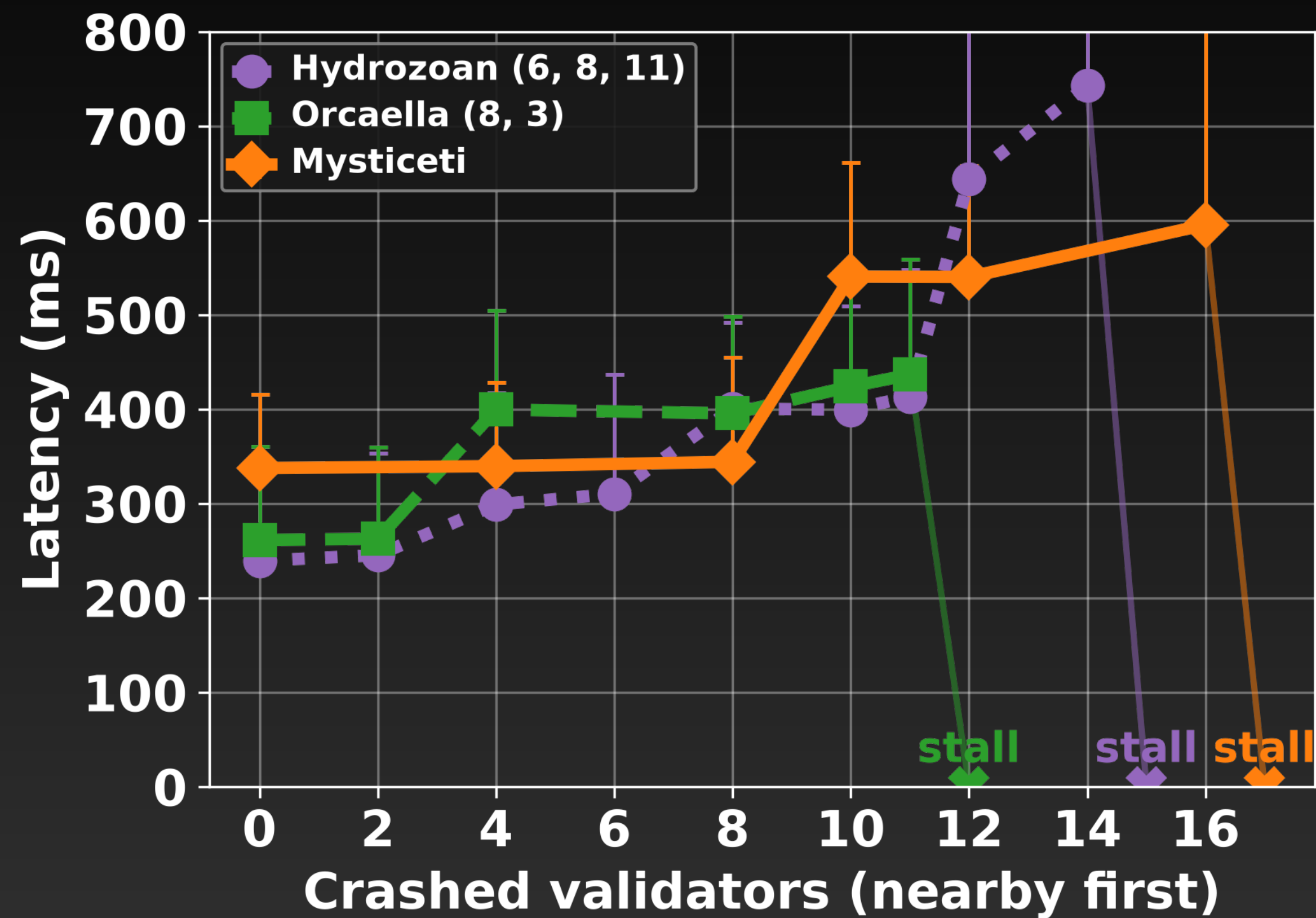


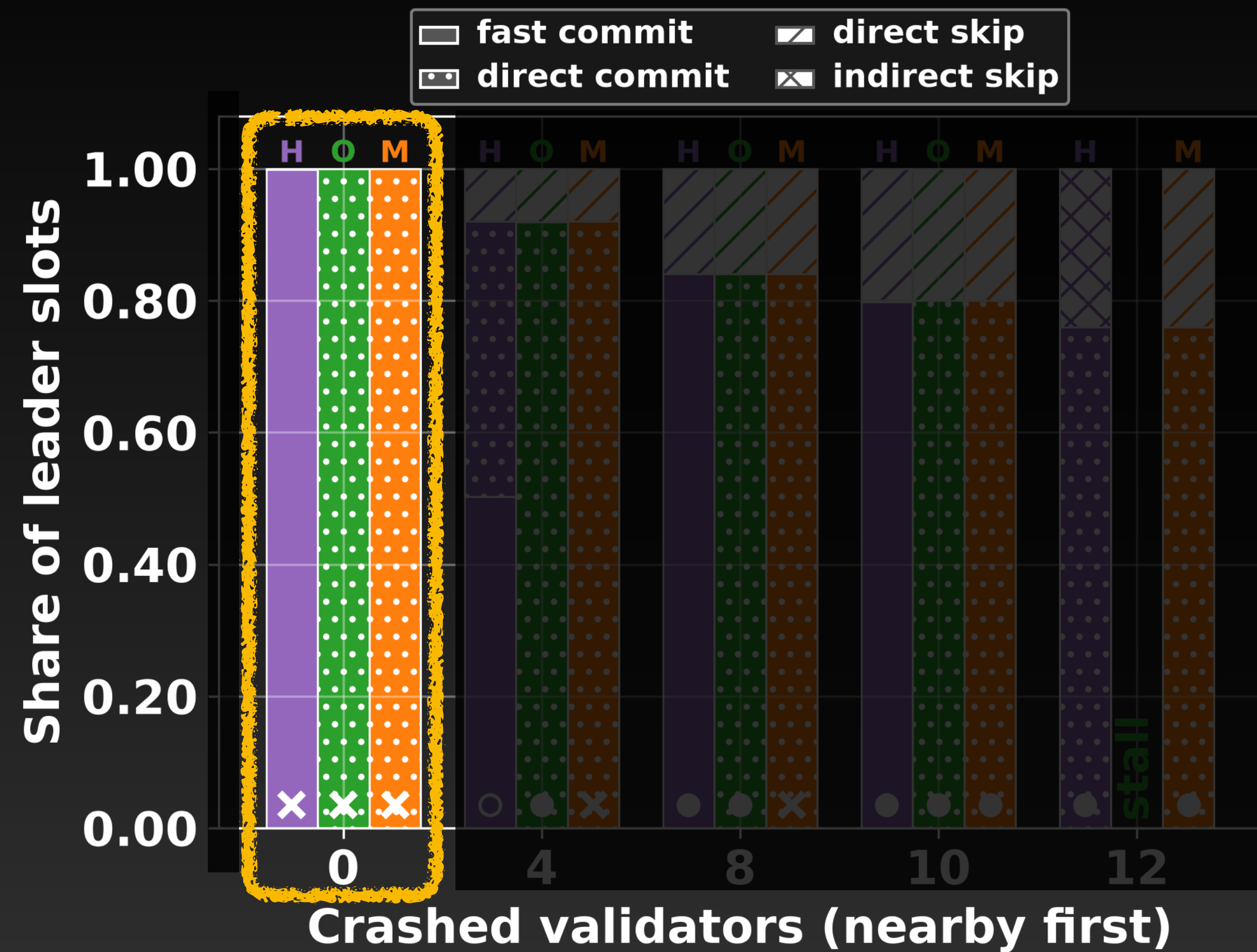
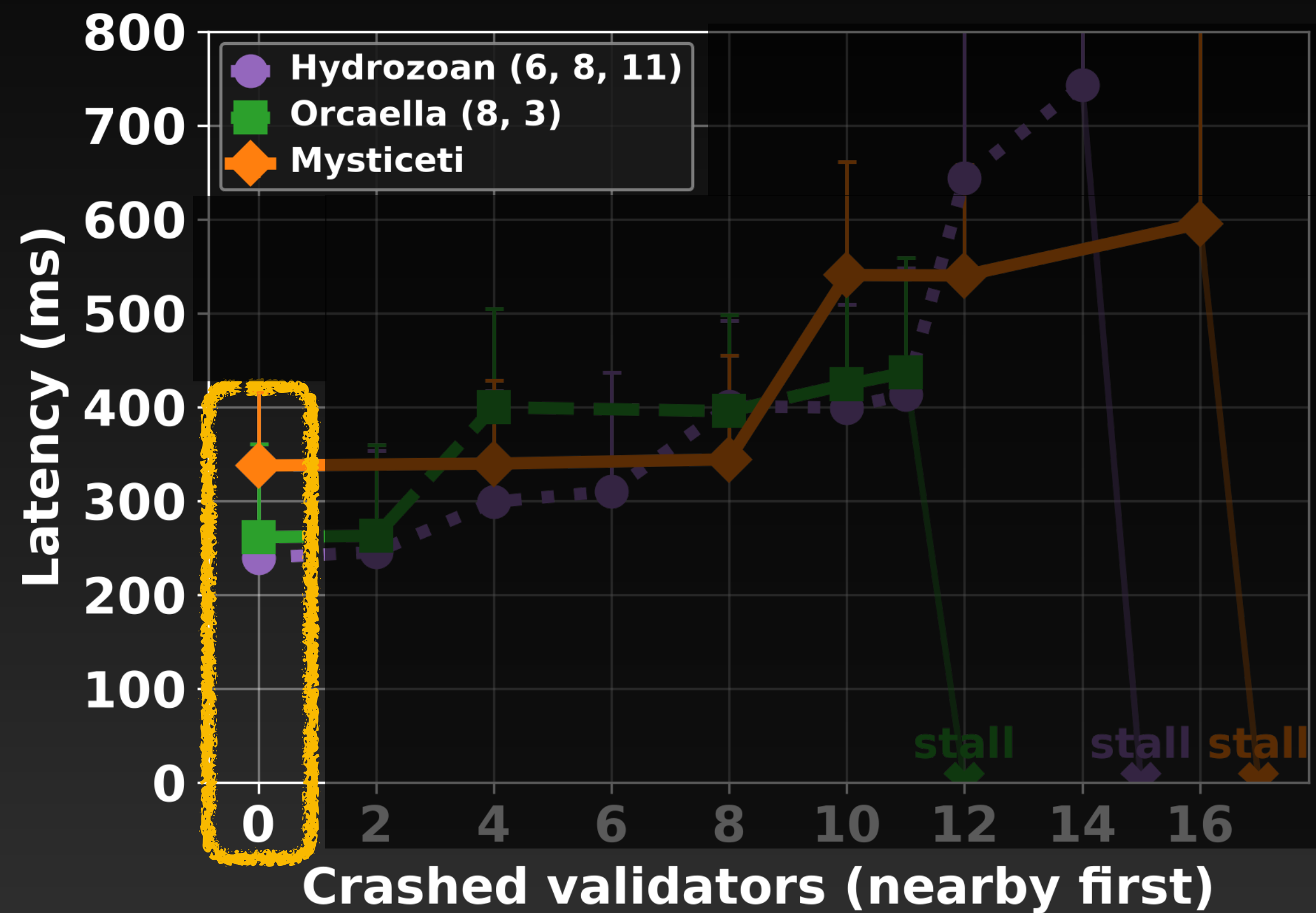


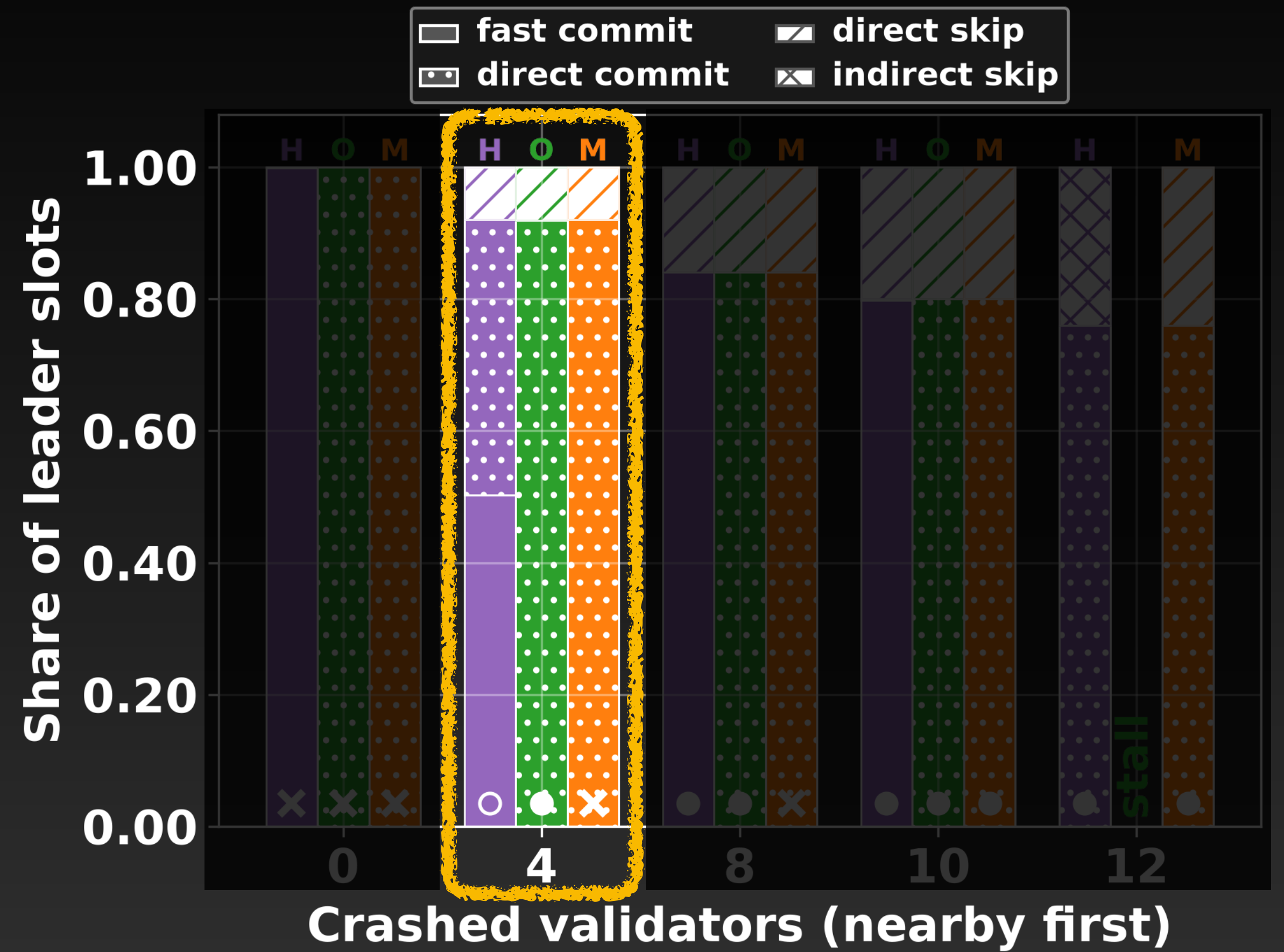
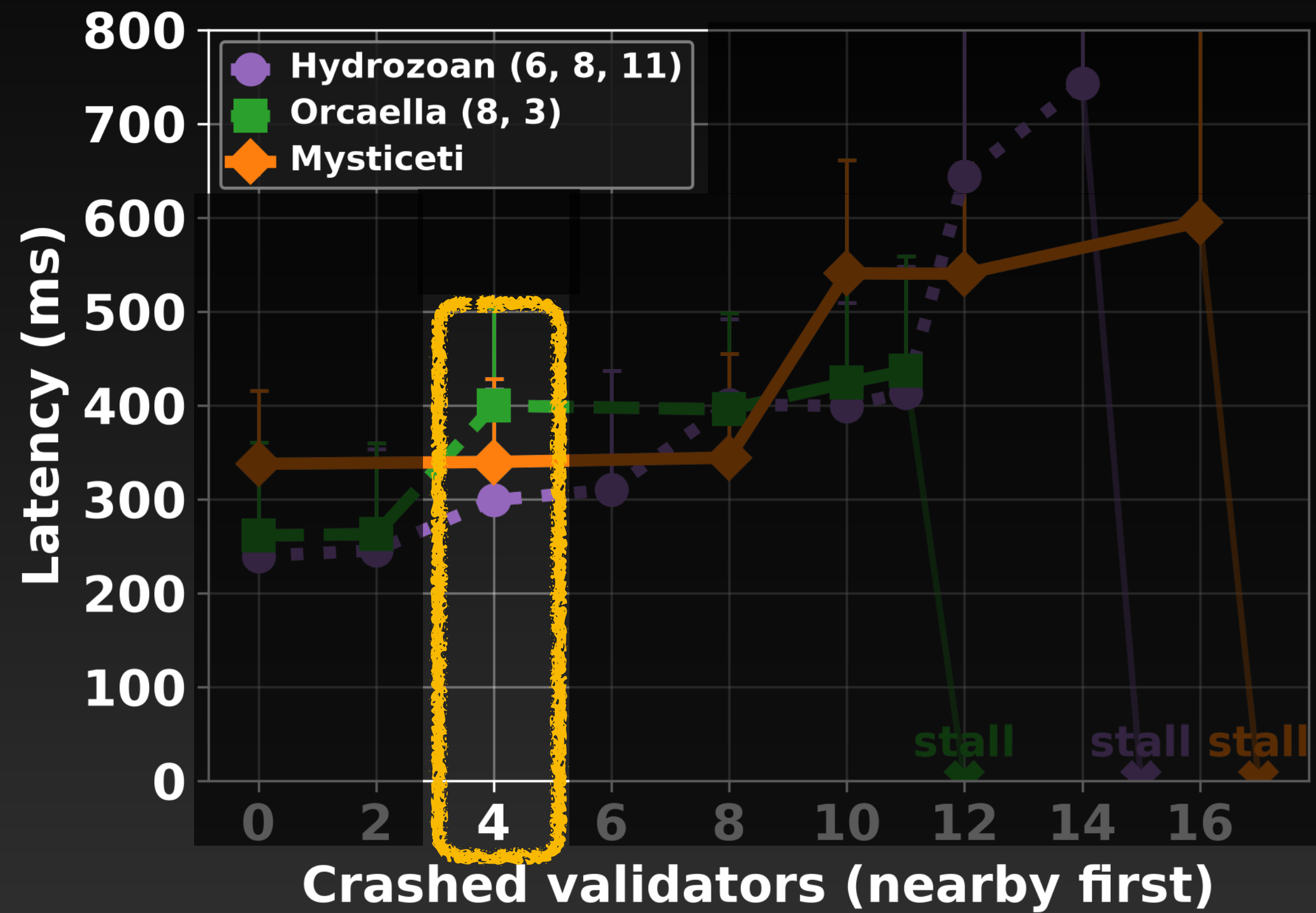
73% of Stake

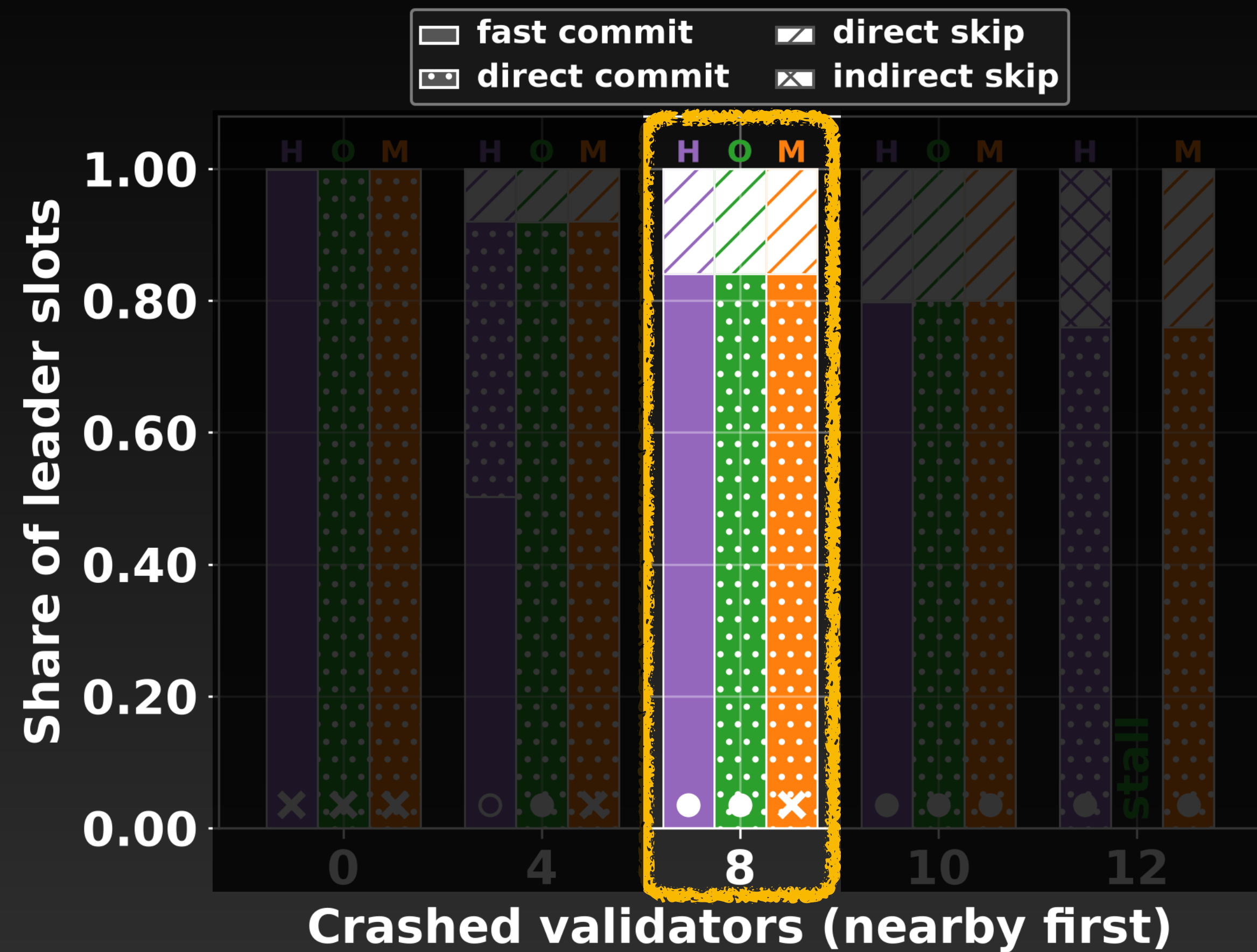
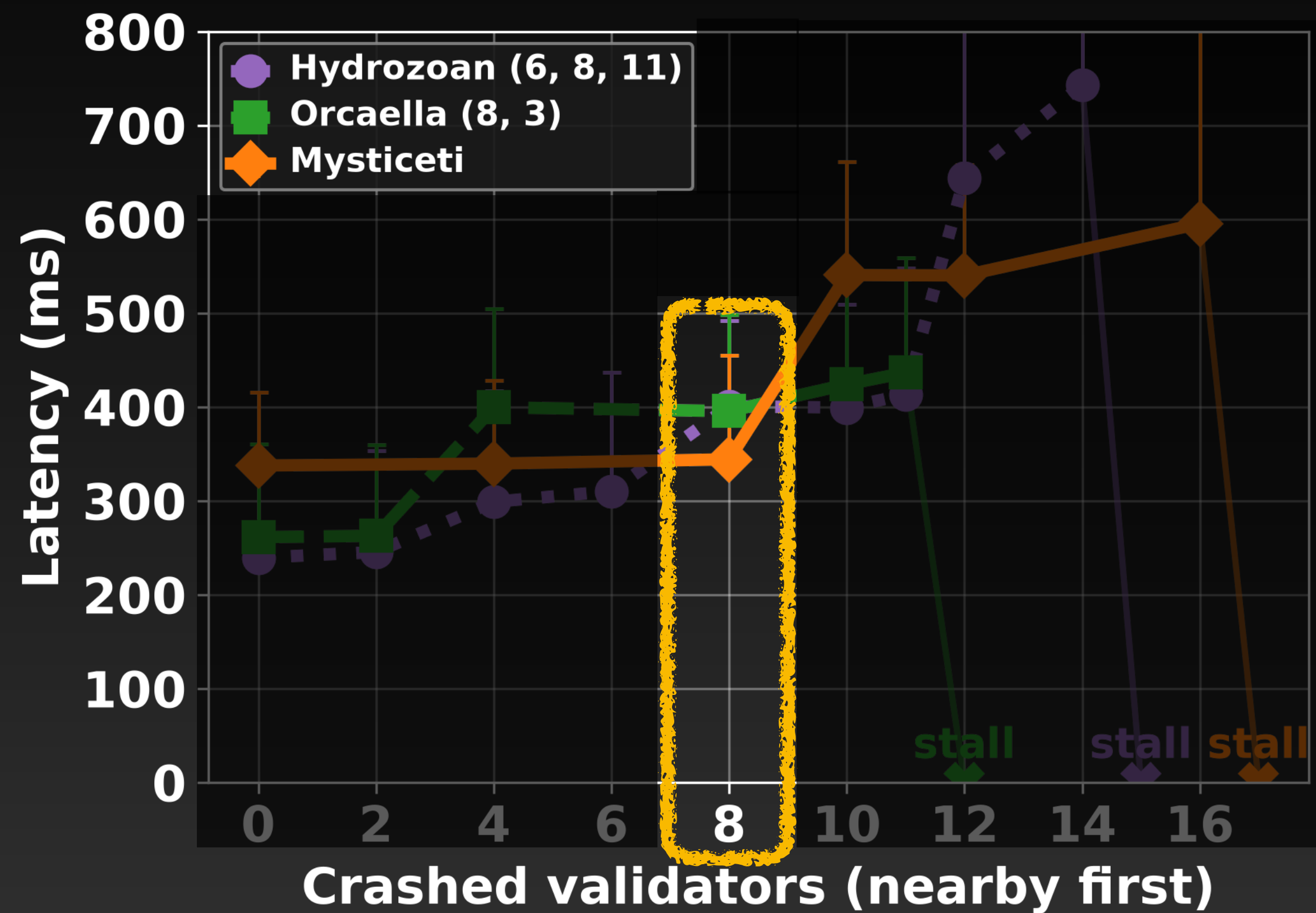


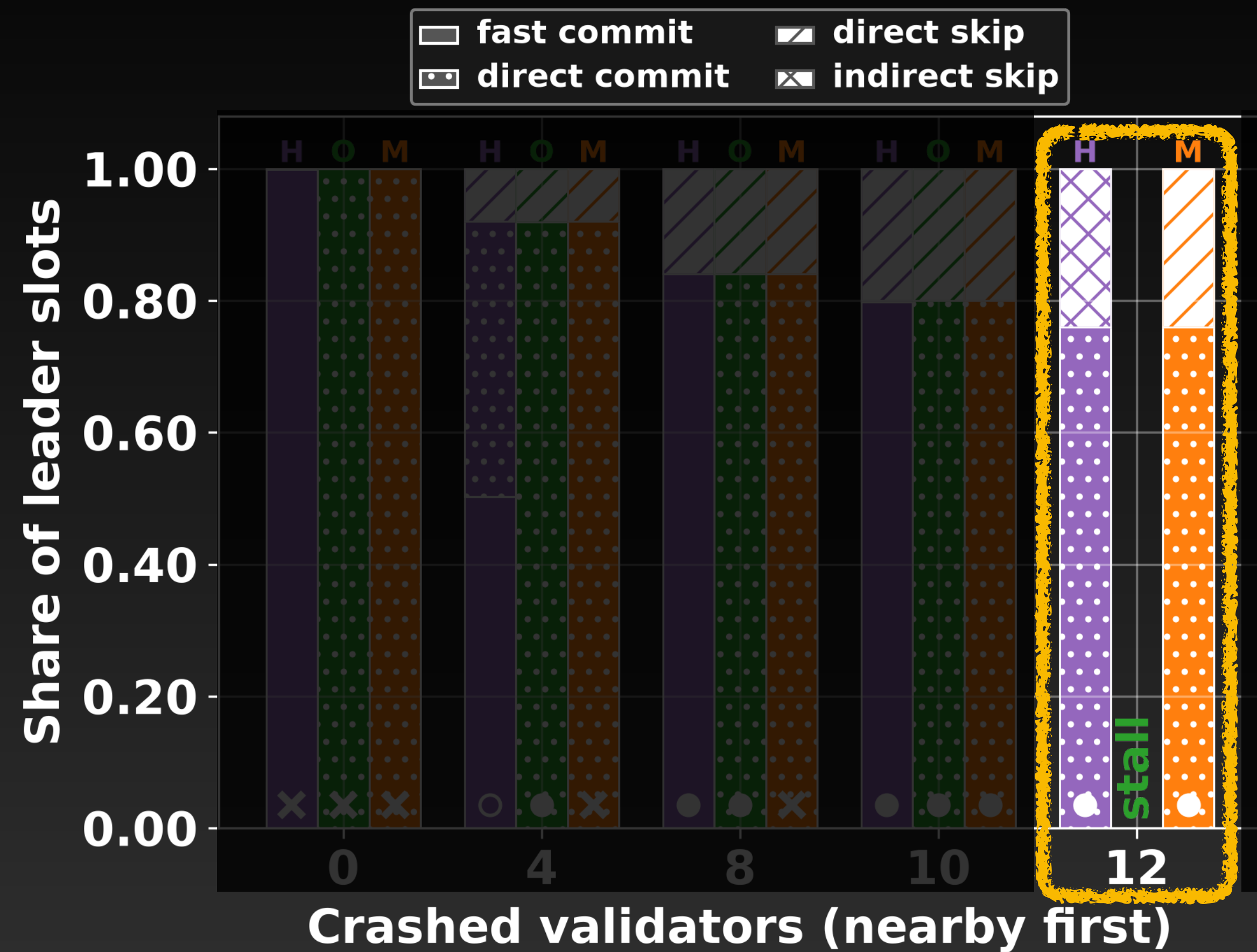
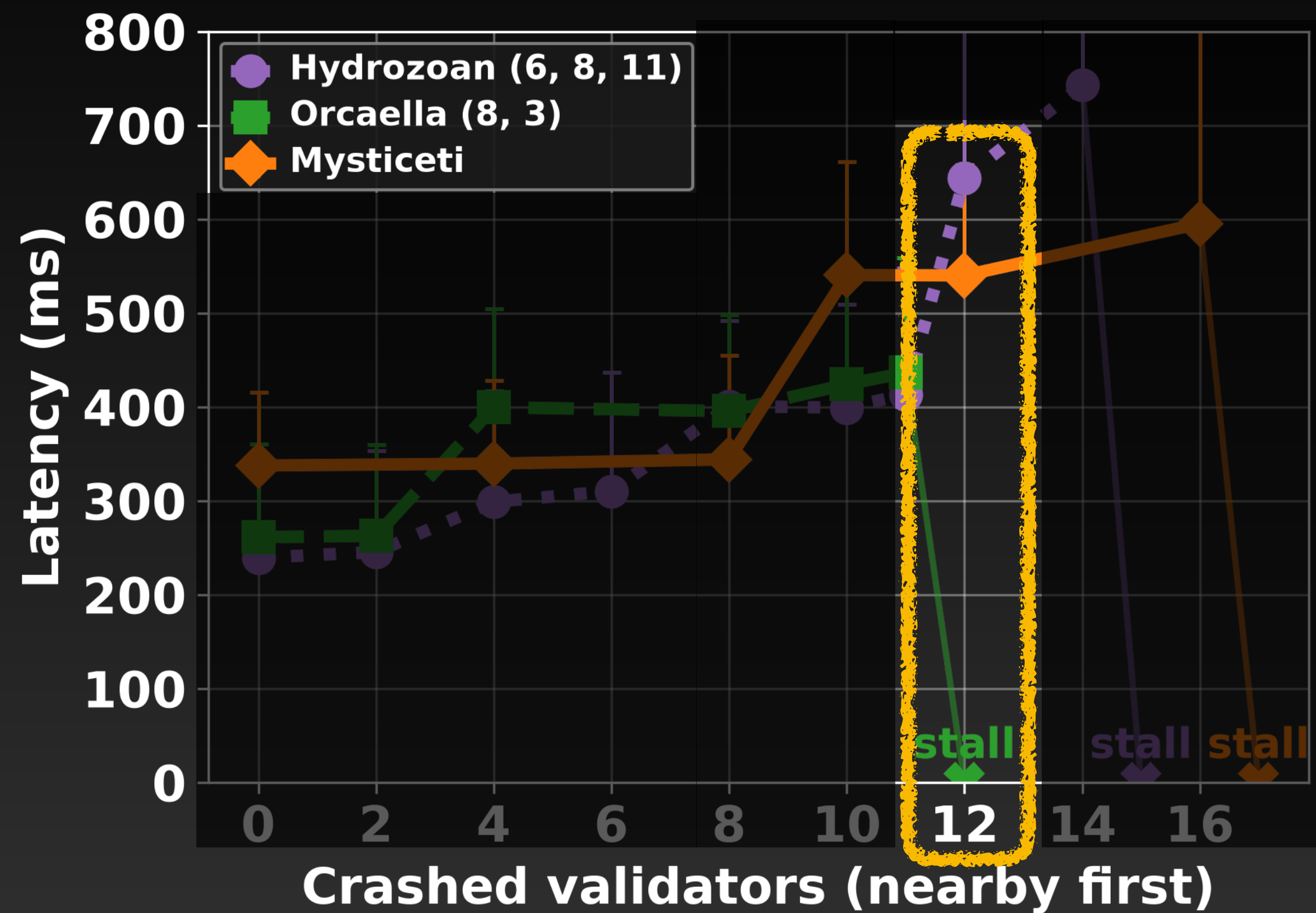
n=50 aws setup

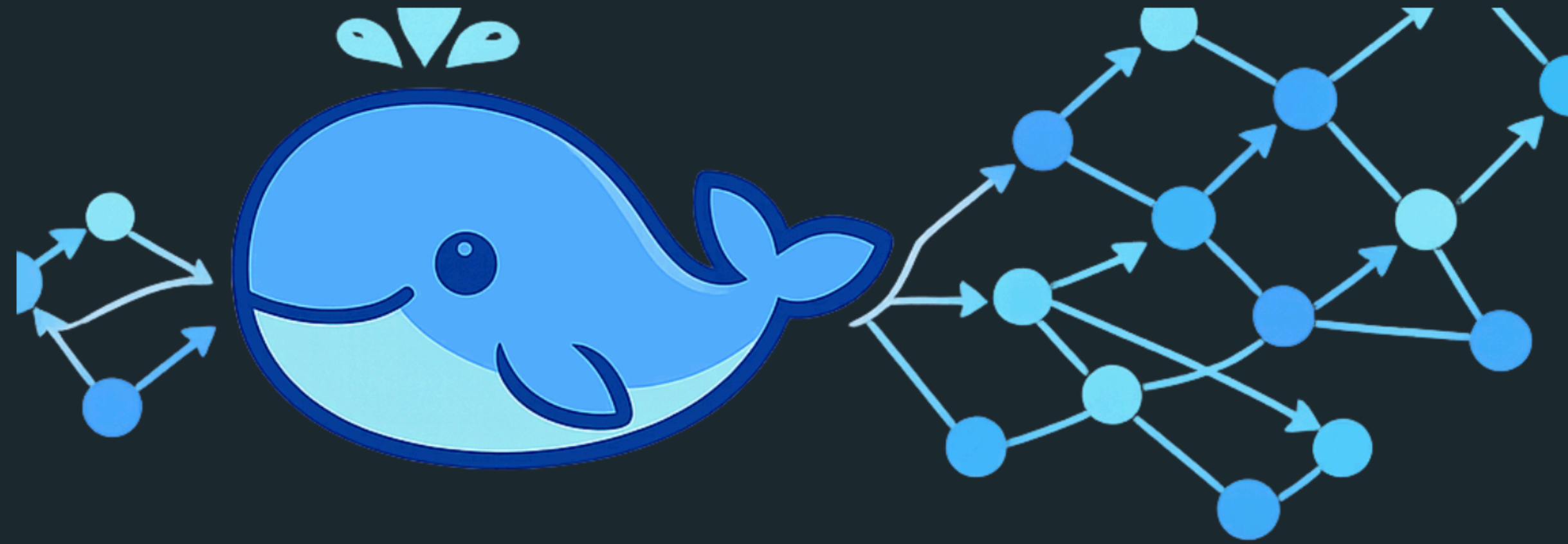












Uncertified DAG Consensus

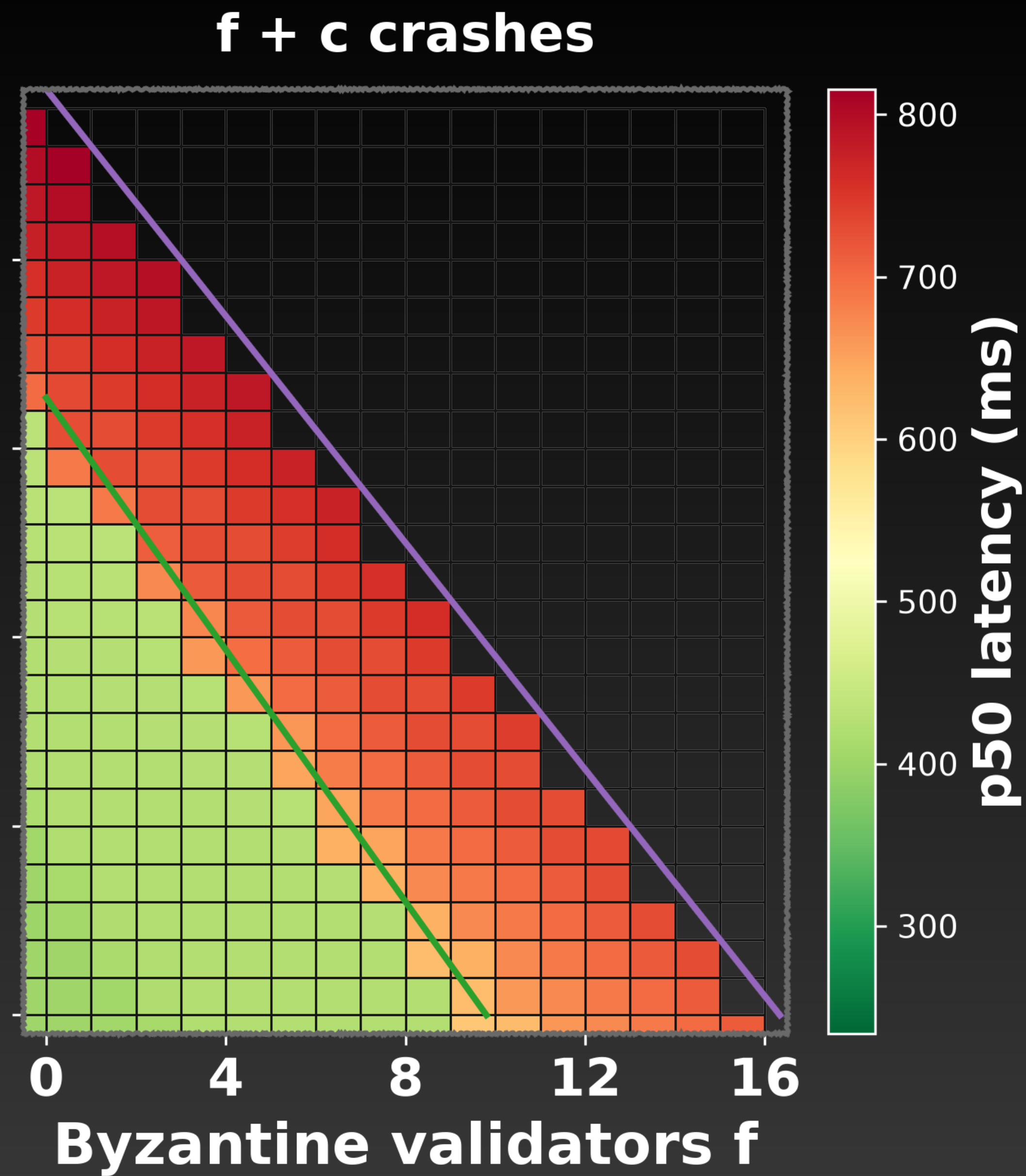
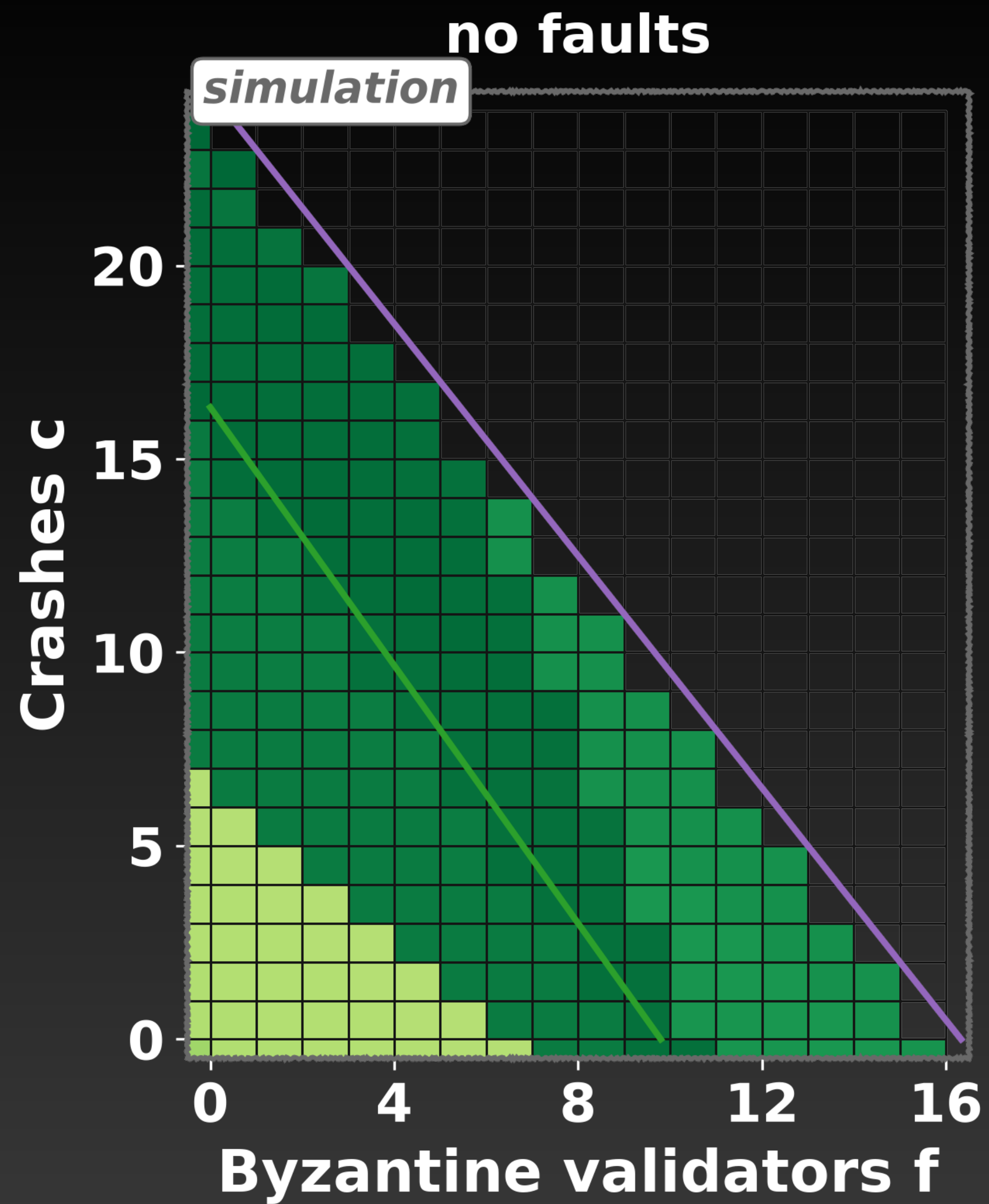
 build **passing**  rustc 1.92+ license Apache

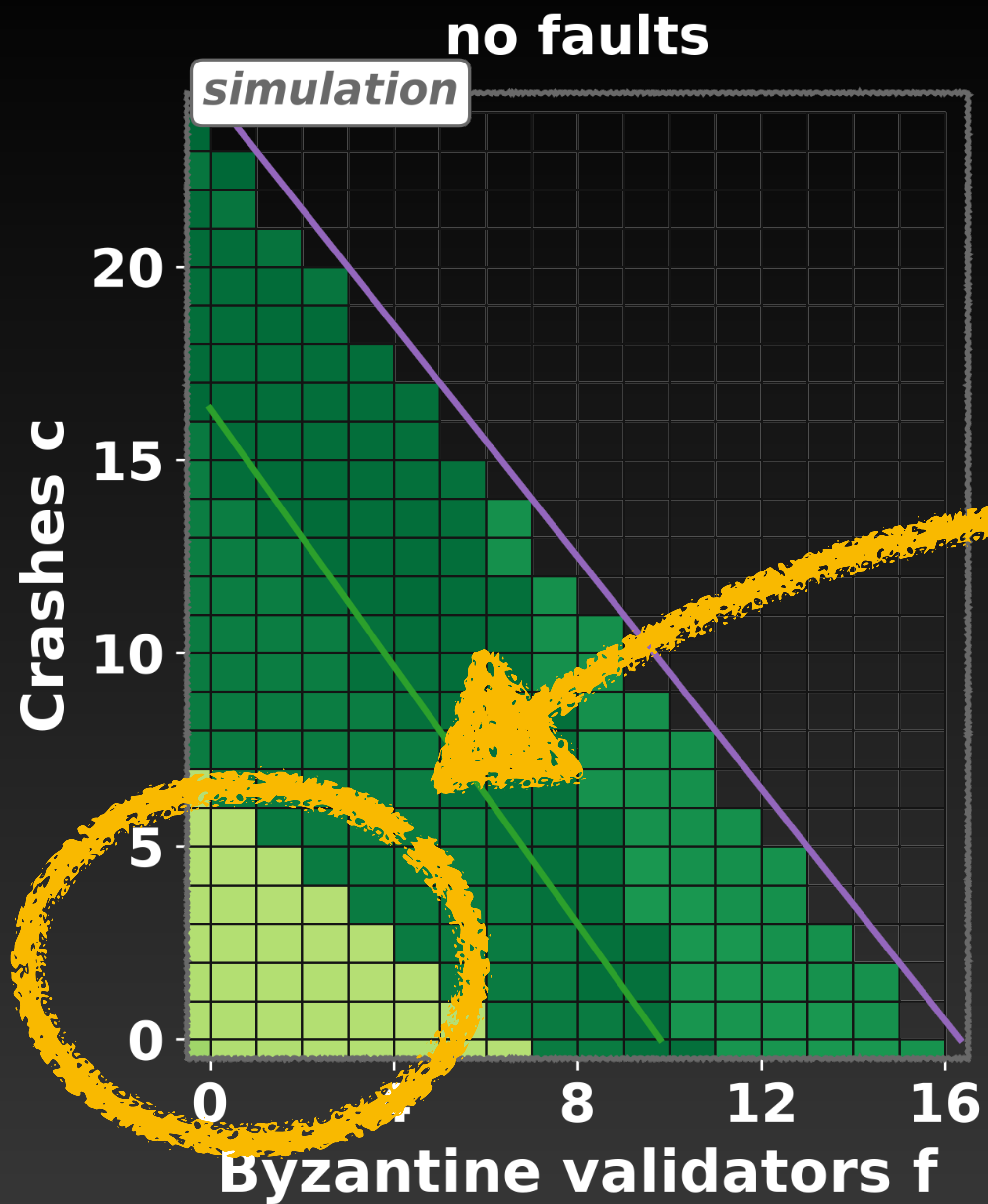
This repository provides reference implementations of several uncertified DAG-based consensus protocols. The codebase is designed to be small, modular, and easy to benchmark and modify. It is not intended for production use, but relies on real networking, cryptography, and a persistent storage. It also ships with a discrete-event simulator for exercising the protocols under custom network and failure scenarios.

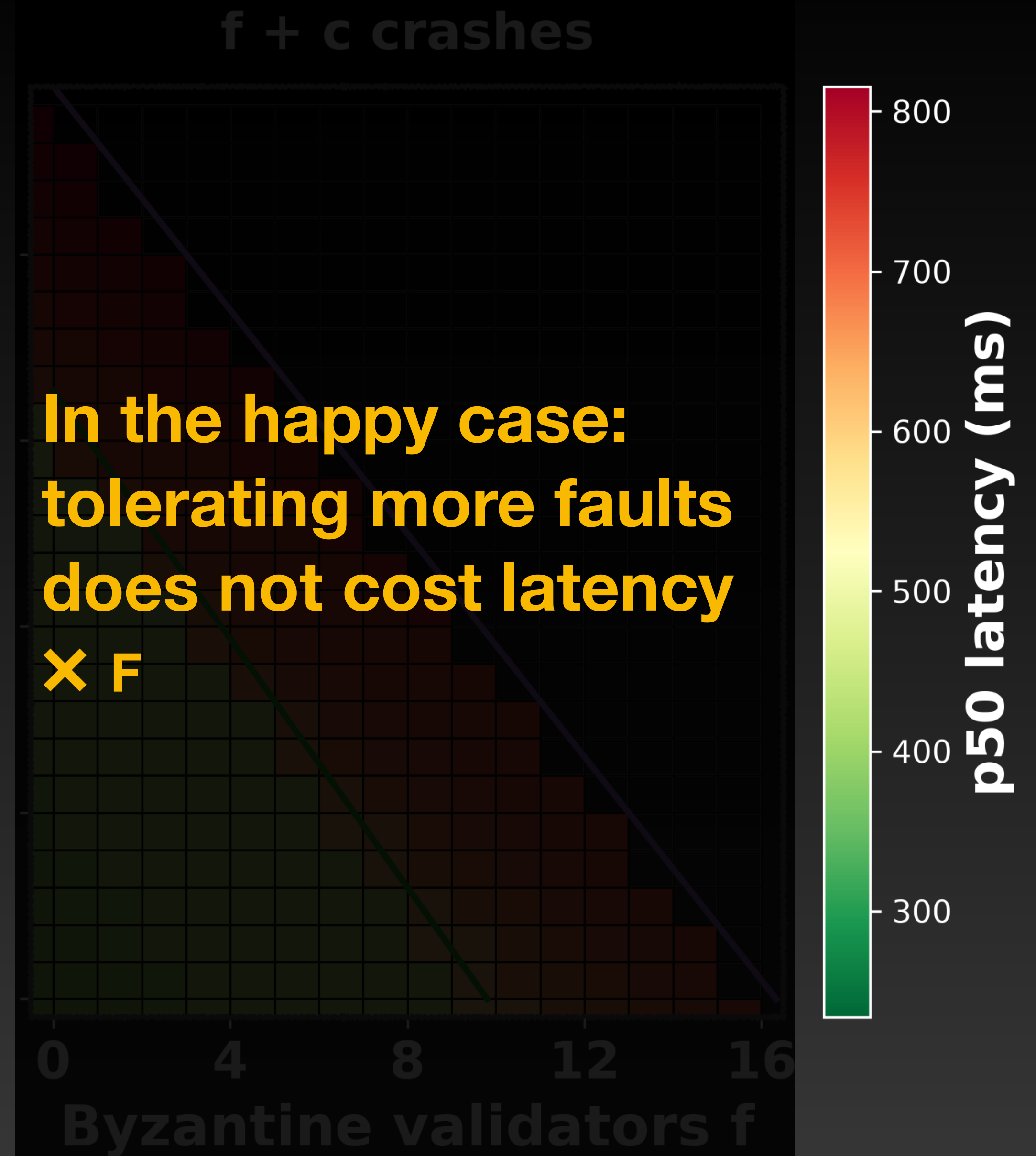
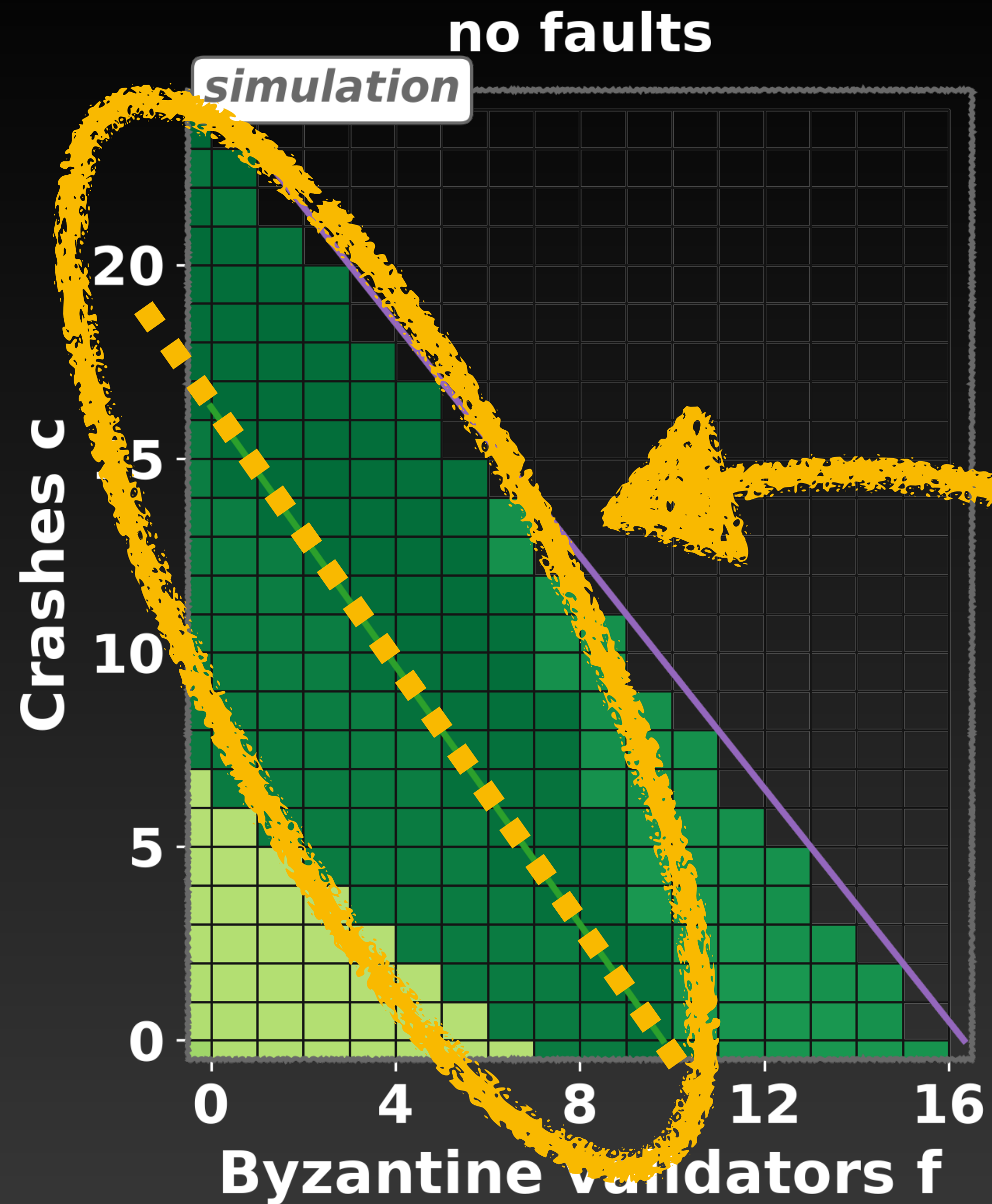
This repository currently supports [Mysticeti](#), [Mahi-Mahi](#), [Blue Bottle](#), [Cordial Miners](#) (both the partially synchronous and asynchronous variants), [Nemo-Nemo](#), [Orcaella](#), and a DAG variant of [Hydrangea](#).

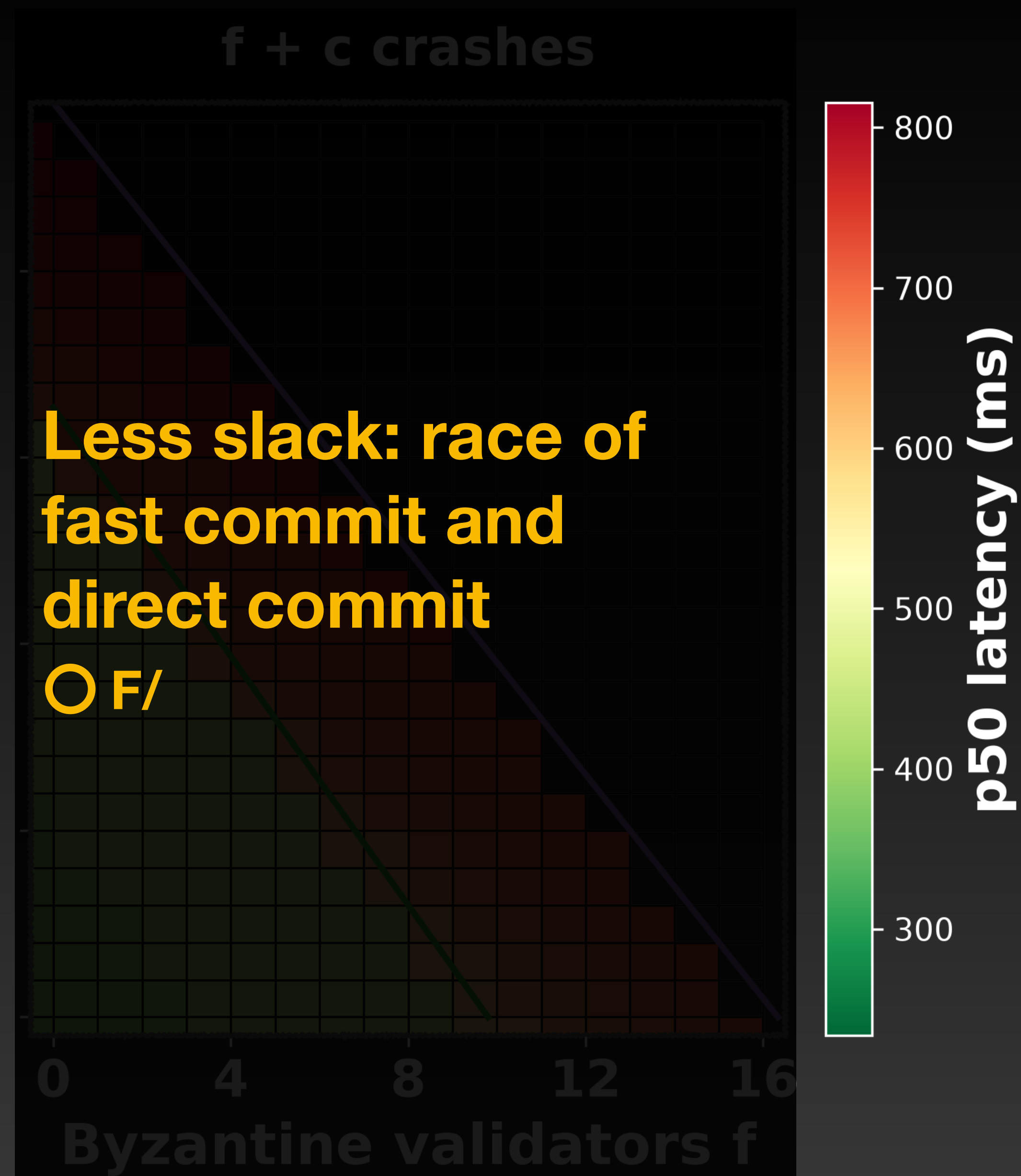
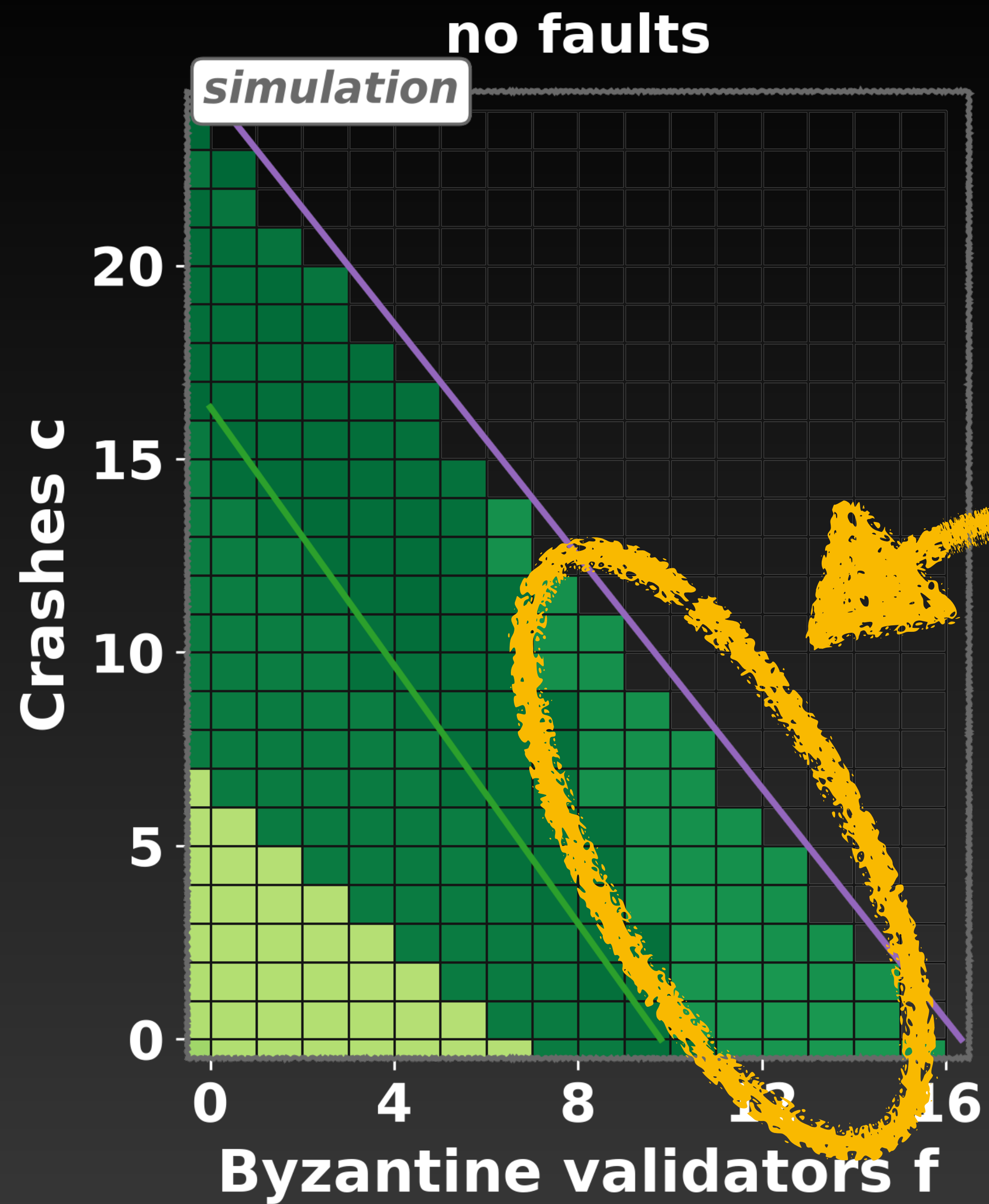
Quick Start

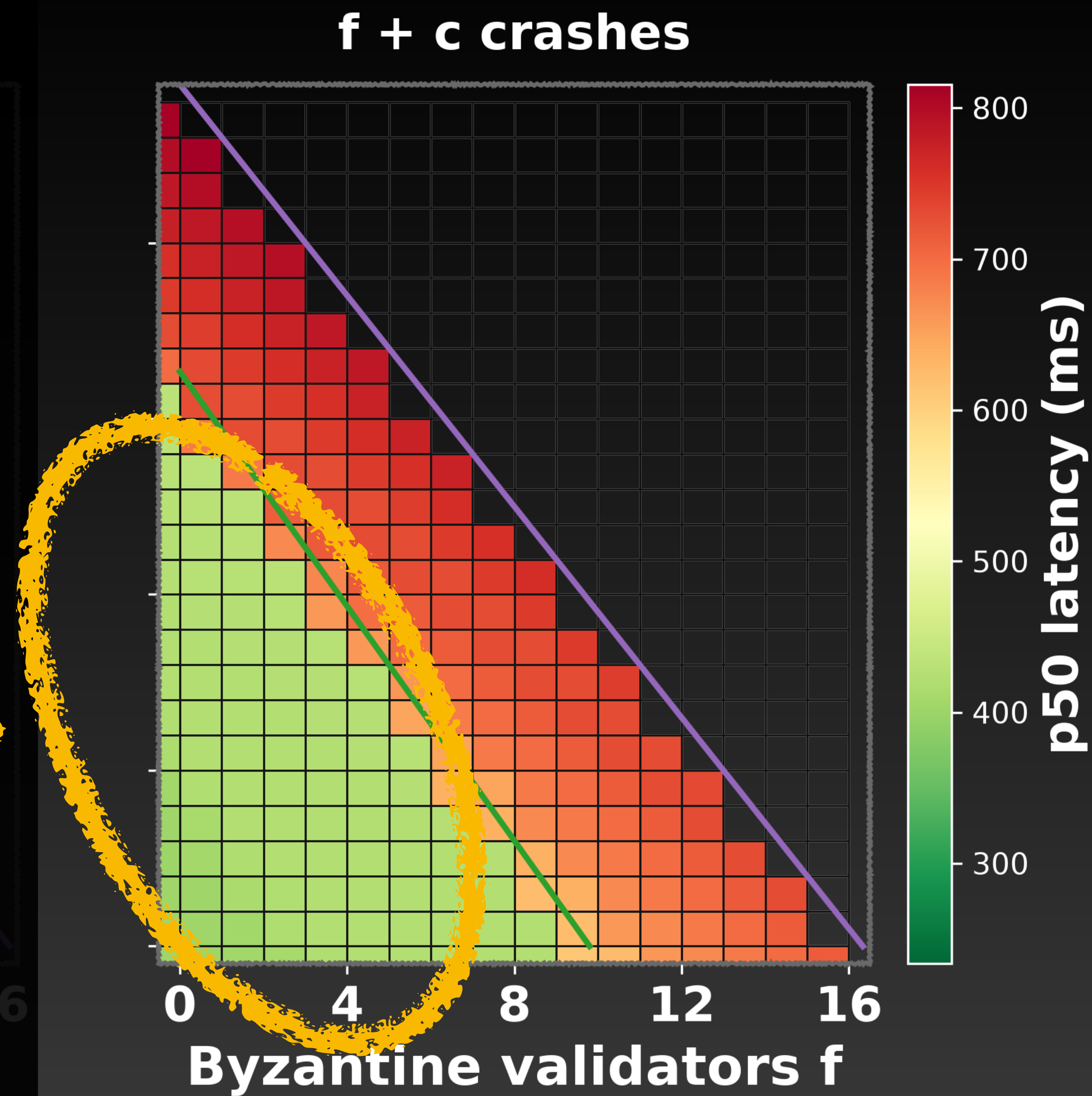
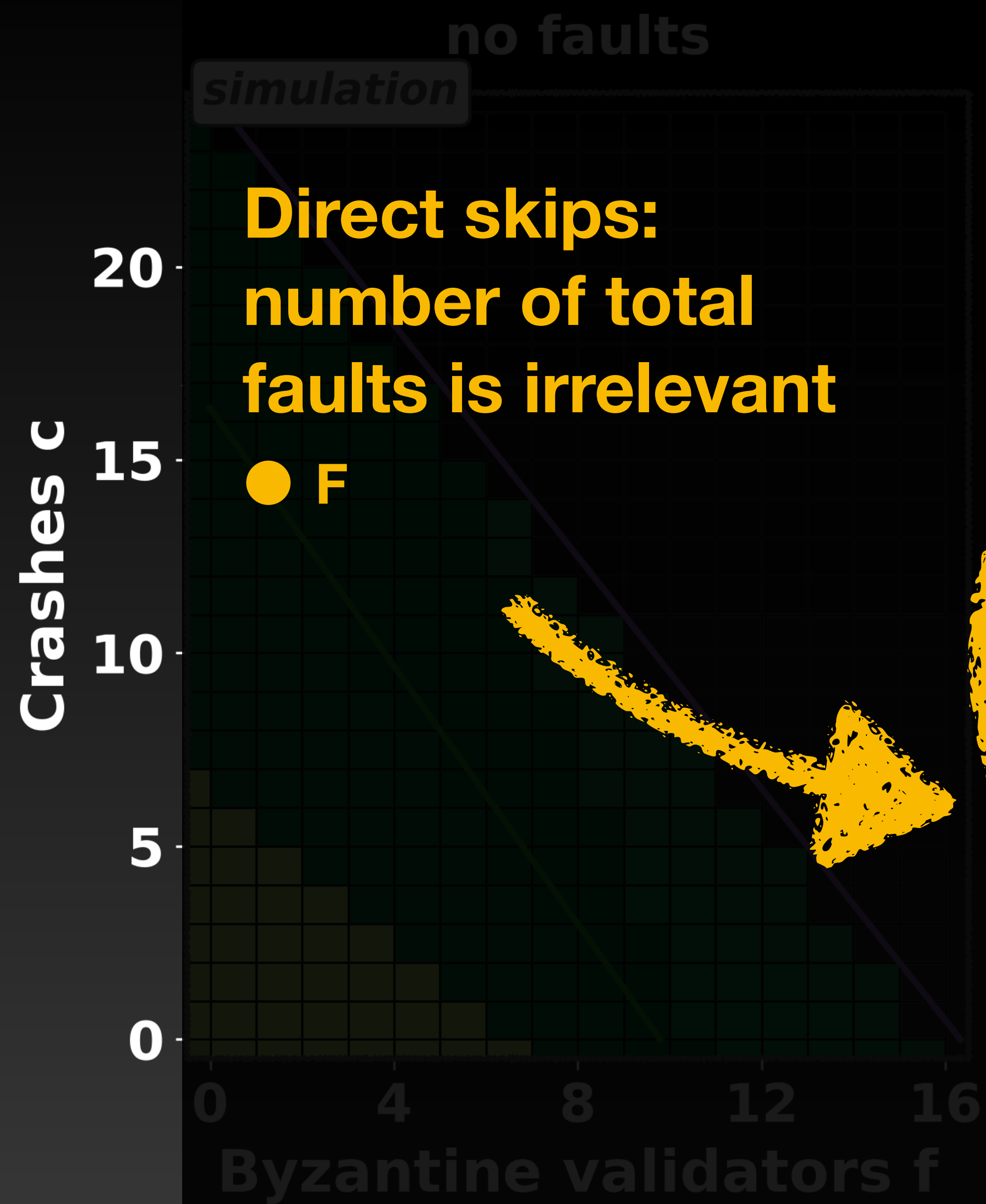
<https://github.com/asonnino/mysticeti>

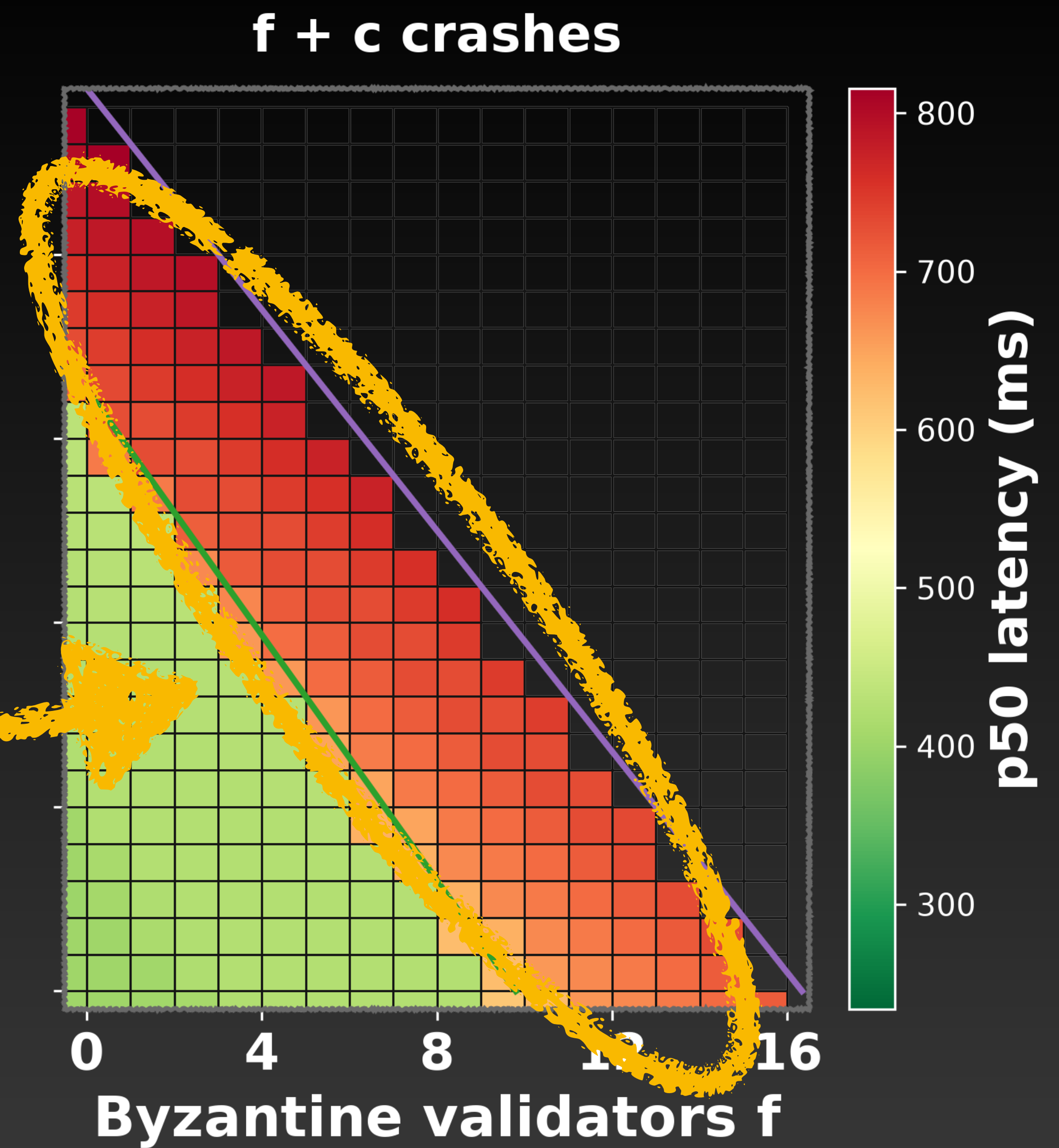


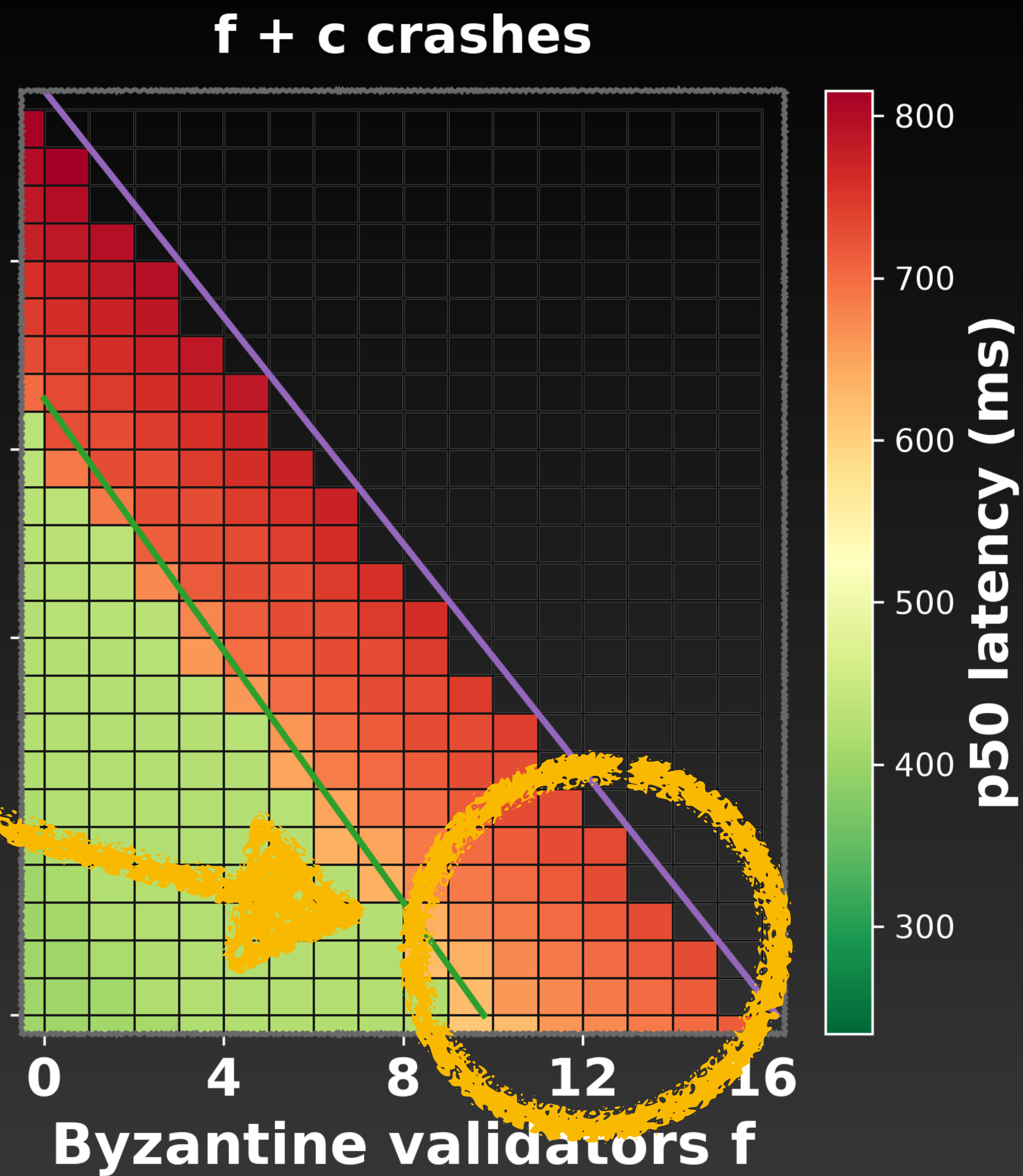
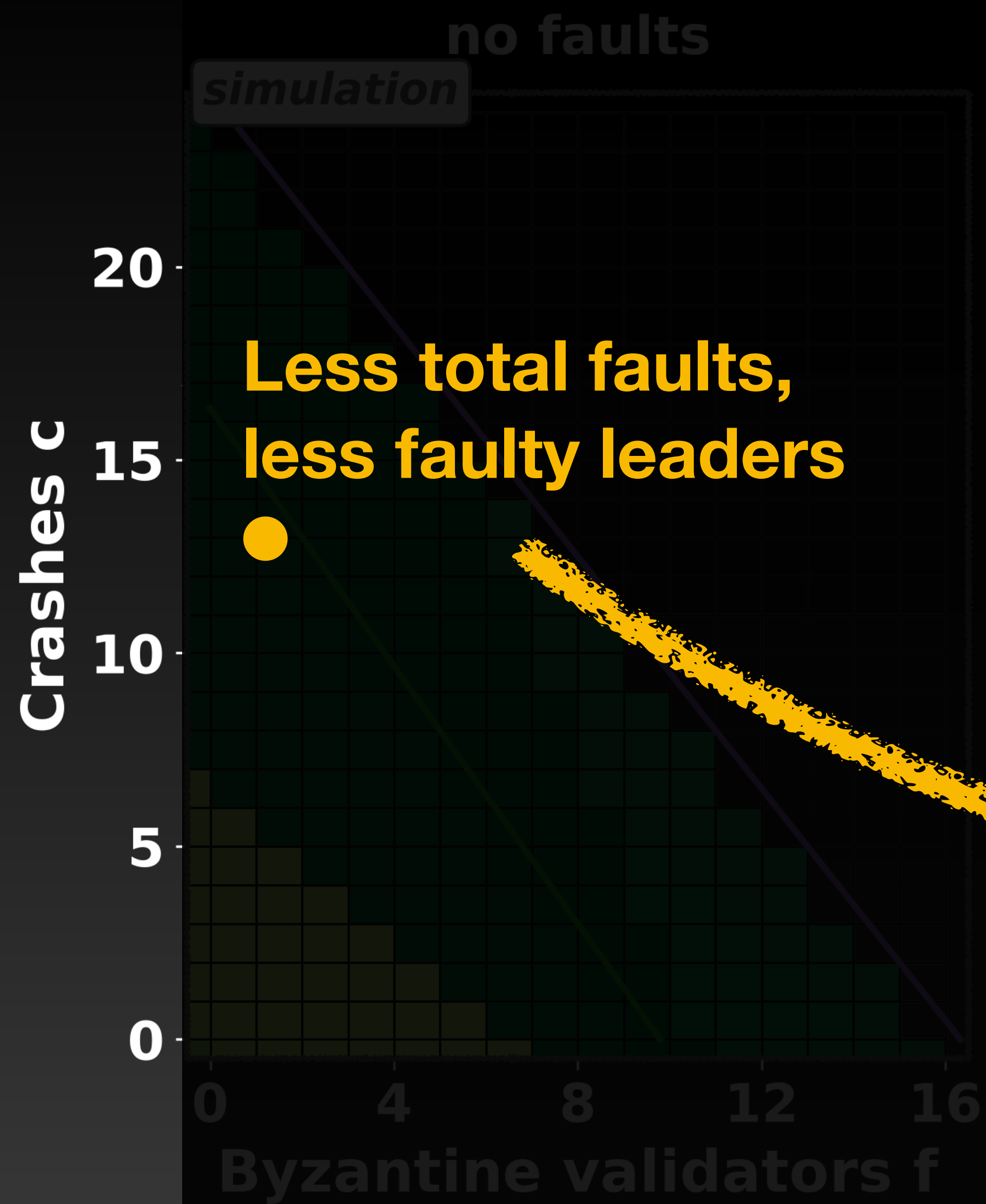












Hydrozoan vs Mysticeti

Fast commit vs tolerant direct skip

Hydrozoan vs Orcaella

Extra 3-round direct commit vs 2-round indirect decision

alberto@mystenlabs.com